Topic No. 76

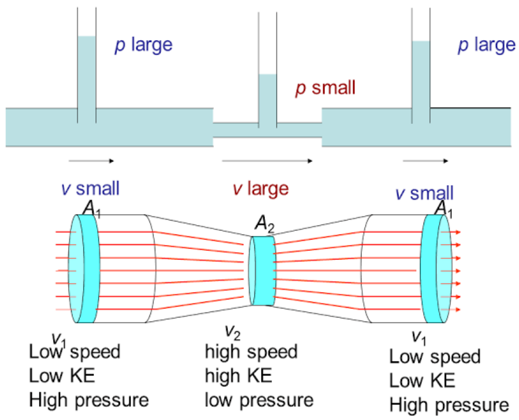
Applications of the Bernoulli's Equation-1

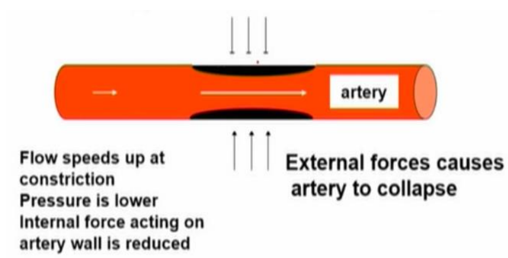
- How can a plane fly?
- How does a perfume spray work?
- What is the venturi effect?
- Why does a cricket ball swing or a baseball curve?

FLUID FLOW : IDEAL FLUID

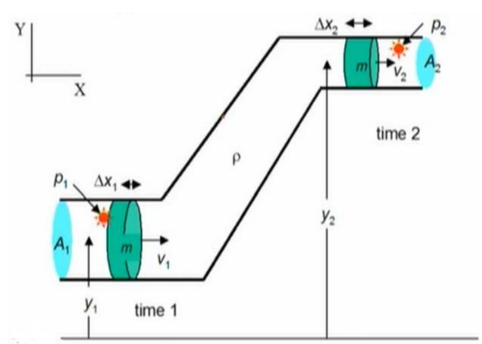
- As the speed of a fluid increases, the pressure in the fluid decreases.

$$P \propto \frac{1}{v}$$





Arteriosclerosis and vascular flutter



- Bernoulli's Equation for any point along a flow tube or streamline

$$p + \frac{1}{2}\rho v^2 + \rho g h = \text{constant}$$

Dimensions:

$$p\left[Pa\right] = \left[Nm^{-2}\right] = \left[Nmm^{-3}\right] = \left[Jm^{-3}\right]$$

$$\frac{1}{2}\rho v^2\left[kgm^{-3}m^2s^{-2}\right] = \left[kgm^{-1}s^{-2}\right] = \left[Nmm^{-3}\right] = \left[Jm^{-3}\right]$$

$$\rho gh\left[kgm^{-3}ms^{-2}m\right] = \left[kgms^{-2}mm^{-3}\right] = \left[Nmm^{-3}\right] = \left[Jm^{-3}\right]$$

Each term has the dimensions of energy/volume or energy density.

$\frac{1}{2}\rho v^2$ KE of bulk motion of fluid

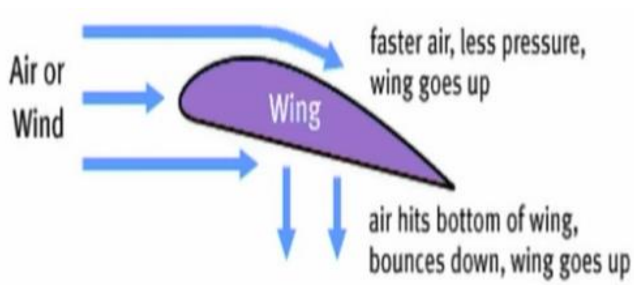
ρgh GPE for location of fluid

p Pressure energy density arising from internal forces within moving fluid (similar to energy stored in a spring)

- Bernoulli's Principle is what allows bird and planes to fly.
- The secret behind flight is under the wings.

Topic No. 77

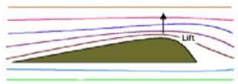
Applications of the Bernoulli's Equation-2



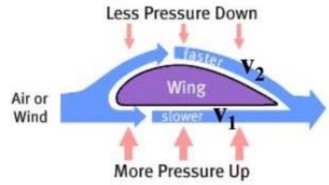
$$\frac{p}{\rho}+\frac{1}{2}v^2+gh=\text{constant}$$

AIRFOIL

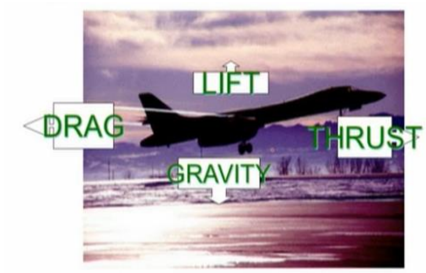
On top: greater air speed and less air pressure

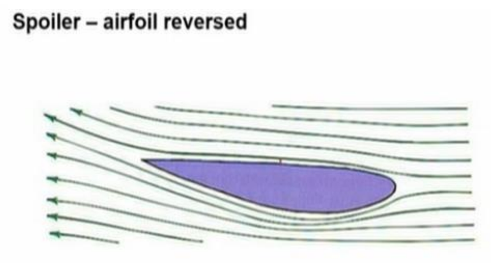


On bottom: less air speed and more air pressure



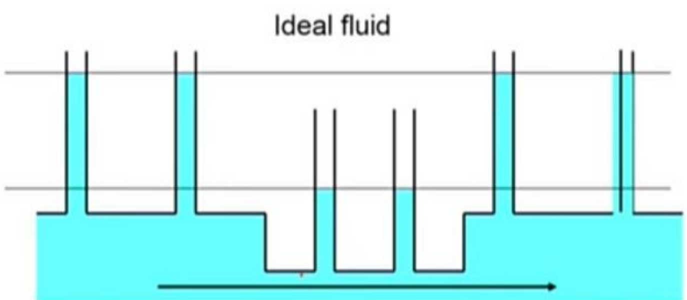
Net force on wing?

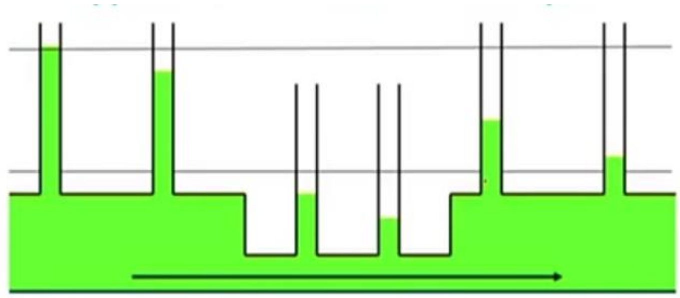




Topic No. 78

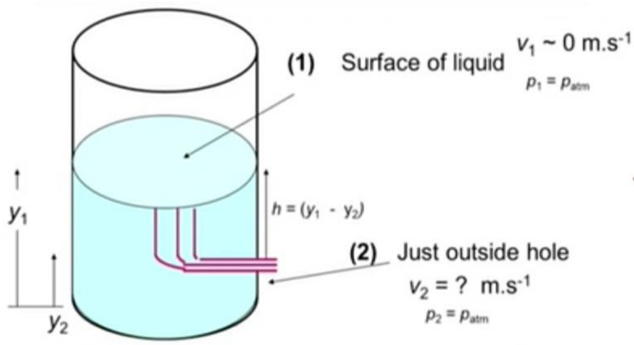
Applications of the Bernoulli's Equation-3





For the viscous fluid there is a gradual decrease in pressure along the pipe.

- Viscosity $\Rightarrow$ frictional forces $\Rightarrow$ energy lost by heating fluid $\Rightarrow$ increase in interval energy of fluid $\Rightarrow$ increase in temperature of fluid.
- Where does the energy come from?
- Equation of continuity still applies $\Rightarrow$ energy cannot come from KE of fluid $\Rightarrow$ energy must be provided from the energy of fluid $\Rightarrow$ pressure drop along tube.



Assume liquid behaves as an ideal fluid and that Bernoulli's equation can be applied

$$p_1 + \tfrac{1}{2}\rho v_1^2 + \rho g y_1 = p_2 + \tfrac{1}{2}\rho v_2^2 + \rho g y_2$$

A small hole is at level (2) and the water level (1) drops slowly $\Rightarrow$ $v_1 = 0$

$$p_1 = p_{atm} \quad , \quad p_2 = p_{atm}$$

$$\rho g y_1 = \tfrac{1}{2}\rho v_2^2 + \rho g y_2$$

$$v_2^2 = 2g(y_1 - y_2) = 2gh \quad , \quad h = (y_1 - y_2)$$

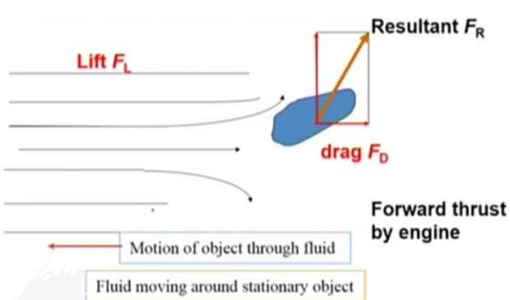
$$v_2 = \sqrt{2gh} \qquad \text{Torricelli formula (1608} - \text{1647)}$$

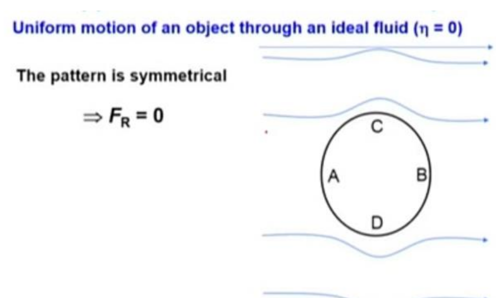
This is the same velocity as a particle falling freely through a height h.

Topic No. 79

Applications of the Bernoulli's Equation-4

FORCES ACTING ON OBJECT MOVING THROUGH FLUID

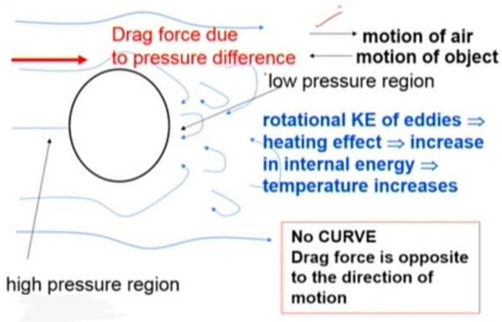


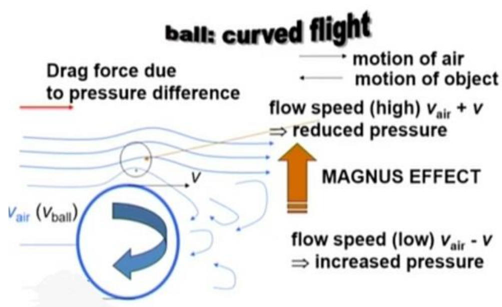


- Why does a cricket ball swing or a baseball curve?
- Drag force

  Frictional drag (viscosity)

  Pressure drag (eddies – lower pressure)





Topic No. 80

Applications of the Bernoulli's Equation-5

$$p + \tfrac{1}{2}\rho v^2 + \rho g y = \text{constant}$$

Dimensions

p [Pa] = [N.m^{-2}] = [N.m.m^{-3}] = [J.m^{-3}]

$\tfrac{1}{2}\rho v^2$ [kg.m^{-3}.m^2.s^{-2}] = [kg.m^{-1}.s^{-2}] = [N.m.m^{-3}] = [J.m^{-3}]

$\rho g h$ [kg.m^{-3} m.s^{-2}. m] = [kg.m.s^{-2}.m.m^{-3}] = [N.m.m^{-3}] = [J.m^{-3}]

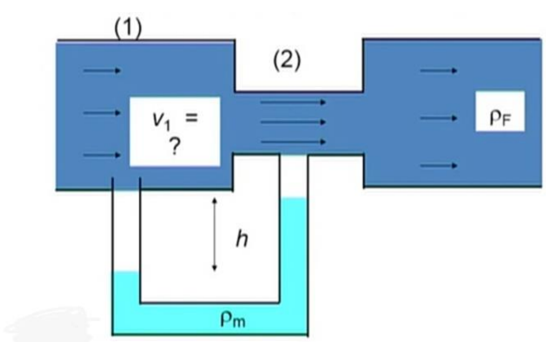
Each term has the dimensions of energy/volume or energy density.

$\frac{1}{2}\rho v^2$ KE of bulk motion of fluid

$\rho g h$ PE for location of fluid

P pressure energy density arising from internal forces within moving fluid (similar to energy stored in a spring)

How do you measure the speed of flow for a fluid?



One method of measuring it is to use a manometer type venture meter.

Assume liquid behaves as an ideal fluid and that Bernoulli's equation can be applied for the flow along a streamline

$$p_1 + \tfrac{1}{2}\rho v_1^2 + \rho g y_1 = p_2 + \tfrac{1}{2}\rho v_2^2 + \rho g y_2$$

$$y_1 = y_2$$

$$p_1 - p_2 = \tfrac{1}{2}\rho_F \left(v_2^2 - v_1^2\right)$$

$$p_1 - p_2 = \rho_m g h$$

$$A_1 v_1 = A_2 v_2 \Rightarrow v_2 = v_1 \left(A_1 / A_2\right)$$

$$\rho_m g h = \tfrac{1}{2}\rho_F \left\{ v_1^2 \left(A_1 / A_2\right)^2 - v_1^2 \right\}$$

$$\rho_m g h = \tfrac{1}{2}\rho_F v_1^2 \left\{ \left(A_1 / A_2\right)^2 - 1 \right\}$$

$$v_1 = \sqrt{\frac{2\rho_m g h}{\rho_F \left\{ \left(A_1 / A_2\right)^2 - 1 \right\}}}$$

Topic No. 81

Applications of the Bernoulli's Equation-6

A large artery in a body has an inner radius of 4.00×10^{-3} m. Blood flows through the artery at the rate of 1.00×10^{-6} m^3s^{-1}. The blood has a viscosity of 2.084×10^{-3} Pa.s and a density of 1.06×10^{3} kgm^{-3}.

Calculate:

(i) The average blood velocity in the artery.

(ii) The pressure drop in a 0.100 m segment of the artery.

(iii) The Reynolds number for the blood flow.

Briefly discuss each of the following:

(iv) The velocity profile across the artery (diagram may be hopeful).

(v) The pressure drop along the segment of the artery.

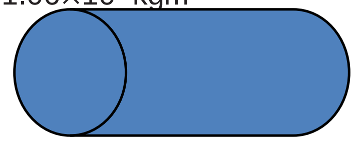
(vi) The significance of the value of the Reynolds number calculated in part (iii).

Solution:

Radius R = 4.00×10^{-3} m

Volume flow rate Q = 1.00×10^{-6} m^3s^{-1}

Viscosity of blood $\eta = 2.084\times10^{-3}$ Pa.s
Density of blood $\rho = 1.06\times10^{3}$ kgm^{-3}



(i) Equation of continuity: $Q = Av$

$$A = \pi R^2 = \pi(4.00\times10^{-3})^2 = 5.03\times10^{-5}\,m^2$$

$$v = \frac{Q}{A} = \frac{1.00\times10^{-6}}{5.03\times10^{-5}} = 1.99\times10^{-2}\,ms^{-1}$$

Topic No. 82

Applications of the Bernoulli's Equation-7

(ii) Poiseuille's Equation:

$$Q = \frac{\Delta P \pi R^4}{8\eta L} \qquad L = 0.100\,m$$

$$\Delta P = \frac{8\eta L Q}{\pi R^4}$$

$$\Delta P = \frac{(8)(2.084\times10^{-3})(0.1)(1.00\times10^{-6})}{(\pi)(4.00\times10^{-3})^4}\,Pa$$

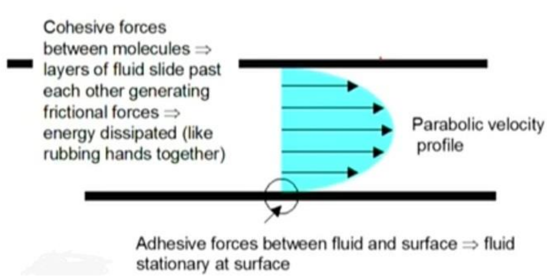
$$\Delta P = 2.07\,Pa$$

(iii) Reynolds Number:

$$R_e = \frac{\rho v L}{\eta} \quad \text{where } L = 2R \text{ (diameter of artery)}$$

$$R_e = \frac{(1.060\times10^{3})(1.99\times10^{-3})(2)(4.00\times10^{-3})}{2.084\times10^{-3}} = 8.1$$

(iv) Parabolic velocity profile: velocity of blood zero at sides of artery
(v) Viscosity ⇒ internal friction ⇒ energy dissipated as thermal energy ⇒ pressure drop along artery
(vi) R_e very small ⇒ laminar flow ($R_e < 2000$)
Flow of a viscous Newtonian fluid through a pipe velocity profile



Topic No. 83

Newton's Laws of Motion-1

OBJECTIVES

- Identify the various kinds of forces and moments acting on a control volume
- Use control volume analysis to determine the forces associated with fluid flow
- Use control volume analysis to determine the moments caused by fluid flow and the torque transmitted

Topic No. 84

Newton's Laws of Motion-2

- Newton's laws:
- Relations between motions of bodies and the forces acting on them.
- Newton's first law: A body at rest remains at rest, and a body in motion remains in motion at the same velocity in a straight path when the net force acting on it is zero.
- Therefore, a body tends to preserve its state of inertia.
- Newton's second law:
- The acceleration of a body is proportional to the net force acting on it and is inversely proportional to its mass.
- Newton's third law:
- When a body exerts a force on a second body, the second body exerts an equal and opposite force on the first.
- Therefore, the direction of an exposed reaction force depends on the body taken as the system.

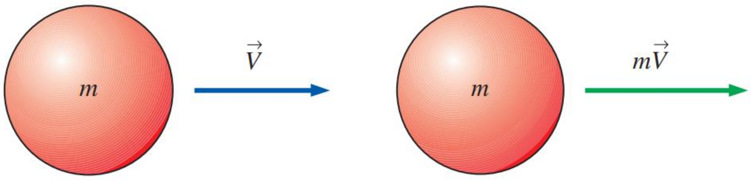
$$\text{Newton's second law:} \quad \vec{F} = m\vec{a} = m\frac{d\vec{V}}{dt} = \frac{d(m\vec{V})}{dt}$$

- Linear momentum or just the momentum of the body: The product of the mass and the velocity of a body.
- Newton's second law is usually referred to as the linear momentum equation.

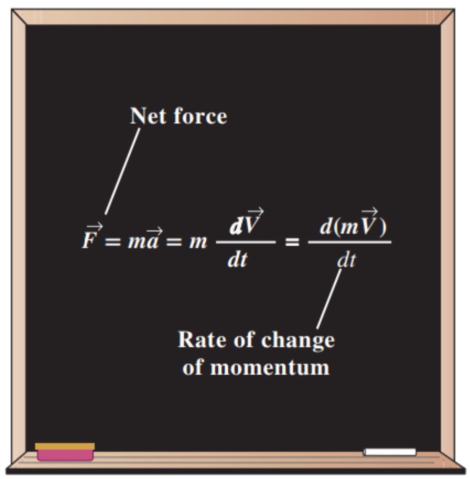
Topic No. 85

Newton's Laws of Motion-3

- Conservation of momentum principle: The momentum of a system remains constant only when the net force acting on it is zero.



- Linear momentum is the product of mass and velocity, and its direction is the direction of velocity.



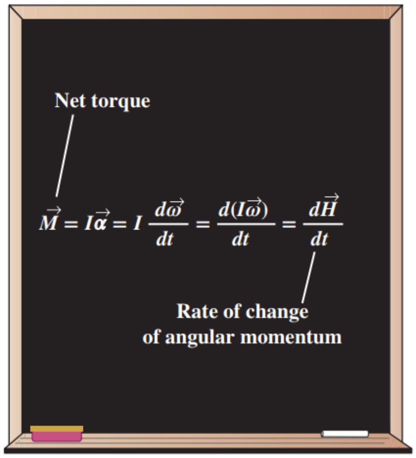
- Newton's second law is also expressed as the rate of change of the momentum of a body is equal to the net force acting on it.

The counterpart of Newton's second law for rotating rigid bodies is expressed as $\vec{M} = I\vec{\alpha}$, where $\vec{M}$ is the net moment or torque applied on the body, I is the moment of inertia of the body about the axis of rotation, and $\vec{\alpha}$ is the angular acceleration. It can also be expressed in terms of the rate of change of angular momentum $d\vec{H}/dt$ as

Angular momentum equation: $$\vec{M} = I\vec{\alpha} = I\frac{d\vec{\omega}}{dt} = \frac{d(I\vec{\omega})}{dt} = \frac{d\vec{H}}{dt}$$

Angular momentum about x-axis: $$M_x = I_x\frac{d\omega_x}{dt} = \frac{dH_x}{dt}$$

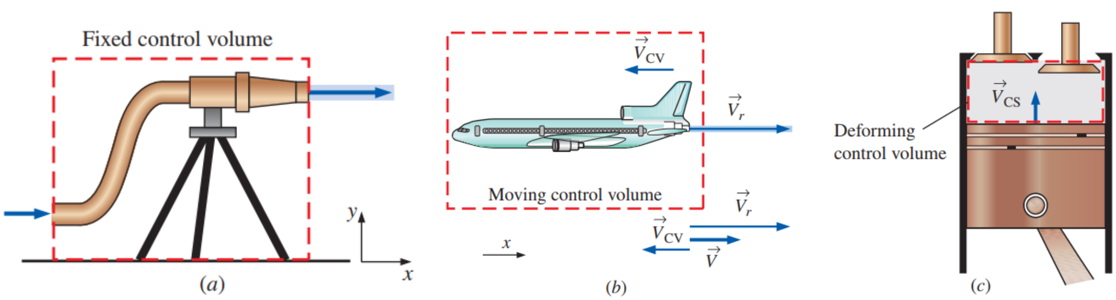
- The conservation of angular momentum principle: The total angular momentum of a rotating body remains constant when the net torque acting on it is zero, and thus the angular momentum of such systems is conserved.



- The rate of change of the angular momentum of a body is equal to the net torque acting on it.

Choosing a Control Volume

- A control volume can be selected as any arbitrary region in space through which fluid flows, and its bounding control surface can be fixed, moving, and even deforming during flow.
- Many flow systems involve stationary hardware firmly fixed to a stationary surface, and such systems are best analyzed using fixed control volumes.
- In deforming control volume, part of the control surface moves relative to other parts.



Examples of (a) fixed, (b) moving, and (c) deforming control volumes.

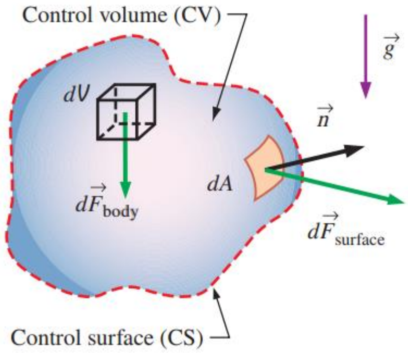
Topic No. 86

Forces Acting On a Control Volume-1

- The forces acting on a control volume consist of **body forces** that act throughout the entire body of the control volume (such as gravity, electric, and magnetic forces) and **surface forces** that act on the control surface (such as pressure and viscous forces and reaction forces at points of contact).

- Only external forces are considered in the analysis.

Total force acting on control volume: $\sum \vec{F} = \sum \vec{F}_{\text{body}} + \sum \vec{F}_{\text{surface}}$



- The total force acting on a control volume is composed of body forces and surface forces; body force is shown on a differential volume element, and surface force is shown on a differential surface element.

- The most common body force is that of **gravity**, which exerts a downward force on every differential element of the control volume.

Gravitational force acting on a fluid element: $d\vec{F}_{\text{gravity}} = \rho \vec{g}\, dV$

Gravitational vector in Cartesian coordinates: $\vec{g} = -g\vec{k}$

Total body force acting on control volume: $\sum \vec{F}_{\text{body}} = \int_{\text{CV}} \rho \vec{g}\, dV = m_{\text{CV}}\vec{g}$

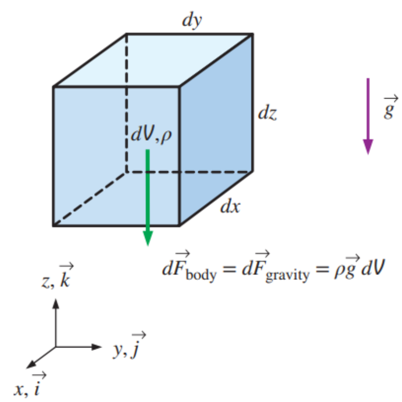
Total force: $\underbrace{\sum \vec{F}}_{\text{total force}} = \underbrace{\sum \vec{F}_{\text{gravity}}}_{\text{body force}} + \underbrace{\sum \vec{F}_{\text{pressure}} + \sum \vec{F}_{\text{viscous}} + \sum \vec{F}_{\text{other}}}_{\text{surface forces}}$

Topic No. 87

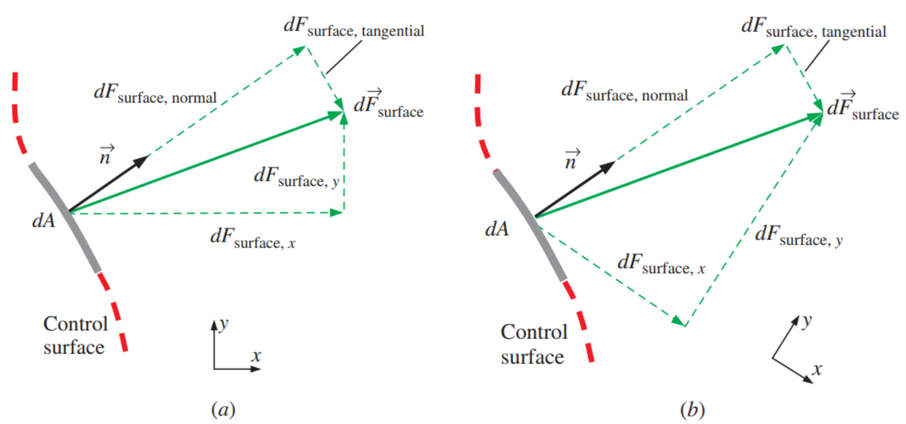
Forces Acting On a Control Volume-2

- Surface forces are not as simple to analyze since they consist of both normal and tangential components.
- **Normal stresses** are composed of pressure (which always acts inwardly normal) and viscous stresses.

- **Shear stresses** are composed entirely of viscous stresses.



- The gravitational force acting on a differential volume element of fluid is equal to its weight; the axes are oriented so that the gravity vector acts downward in the negative z-direction.



- When coordinate axes are rotated (*a*) to (*b*), the components of the surface force change, even though the force itself remains the same; only two dimensions are shown here.

Surface force acting on a differential surface element: $\quad d\vec{F}_{surface} = \sigma_{ij} \cdot \vec{n}\, dA$

Total surface force acting on control surface: $\quad \sum \vec{F}_{surface} = \int_{CS} \sigma_{ij} \cdot \vec{n}\, dA$

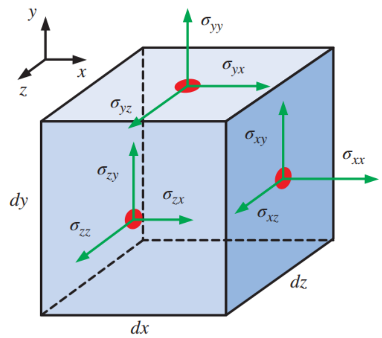
$$\sum \vec{F} = \sum \vec{F}_{body} + \sum \vec{F}_{surface} = \int_{CV} \rho\vec{g}\, dV + \int_{CS} \sigma_{ij} \cdot \vec{n}\, dA$$

Total force:

$$\underbrace{\sum \vec{F}}_{\text{total force}} = \underbrace{\sum \vec{F}_{gravity}}_{\text{body force}} + \underbrace{\sum \vec{F}_{pressure} + \sum \vec{F}_{viscous} + \sum \vec{F}_{other}}_{\text{surface forces}}$$

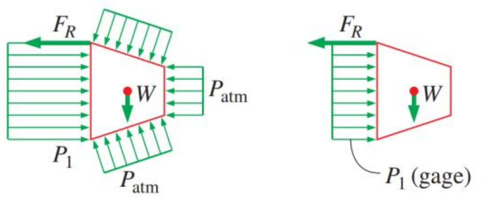
Topic No. 88

Forces Acting On a Control Volume-3



- Components of the stress tensor in Cartesian coordinates on the right, top, and front faces.

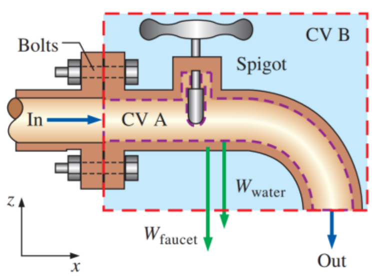
- A common simplification in the application of Newton's laws of motion is to subtract the atmospheric pressure and work with gage pressures.
- This is because atmospheric pressure acts in all directions, and its effect cancels out in every direction.
- This means we can also ignore the pressure forces at outlet sections where the fluid is discharged to the atmosphere since the discharge pressures in such cases are very near atmospheric pressure at subsonic velocities.



With atmospheric pressure considered

With atmospheric pressure cancelled out

- Atmospheric pressure acts in all directions, and thus it can be ignored when performing force balances since its effect cancels out in every direction.



Cross section through a faucet assembly, illustrating the importance of choosing a control volume wisely; CV B is much easier to work with than CV A.

Topic No. 89

The Linear Momentum Equations

Newton's second law for a system of mass *m* subjected to net force $\Sigma\vec{F}$ is expressed as

$$\sum \vec{F} = m\vec{a} = m\frac{d\vec{V}}{dt} = \frac{d}{dt}(m\vec{V})$$

where $m\vec{V}$ is the **linear momentum** of the system. Noting that both the density and velocity may change from point to point within the system, Newton's second law can be expressed more generally as

$$\sum \vec{F} = \frac{d}{dt}\int_{\text{sys}} \rho\vec{V}\,dV$$

where $\rho\vec{V}\,dV$ is the momentum of a differential element dV, which has mass $\delta m = \rho\, dV$.

- Newton's second law can be stated as the sum of all external forces acting on a system is equal to the time rate of change of linear momentum of the system.
- This statement is valid for a coordinate system that is at rest or moves with a constant velocity, called an inertial coordinate system or inertial reference frame.

- The Reynolds transport theorem is expressed for linear momentum as

$$\frac{d(m\vec{V})_{\text{sys}}}{dt} = \frac{d}{dt}\int_{\text{CV}} \rho\vec{V}\,dV + \int_{\text{CS}} \rho\vec{V}\,(\vec{V}_r\cdot\vec{n})\,dA$$

General: $$\sum \vec{F} = \frac{d}{dt}\int_{\text{CV}} \rho\vec{V}\,dV + \int_{\text{CS}} \rho\vec{V}(\vec{V}_r\cdot\vec{n})\,dA$$

$$\begin{pmatrix}\text{The sum of all}\\ \text{external forces}\\ \text{acting on a CV}\end{pmatrix} = \begin{pmatrix}\text{The time rate of change}\\ \text{of the linear momentum}\\ \text{of the contents of the CV}\end{pmatrix} + \begin{pmatrix}\text{The net flow rate of}\\ \text{linear momentum out of the}\\ \text{control surface by mass flow}\end{pmatrix}$$

Fixed CV: $$\sum \vec{F} = \frac{d}{dt}\int_{\text{CV}} \rho\vec{V}\,dV + \int_{\text{CS}} \rho\vec{V}(\vec{V}\cdot\vec{n})\,dA$$

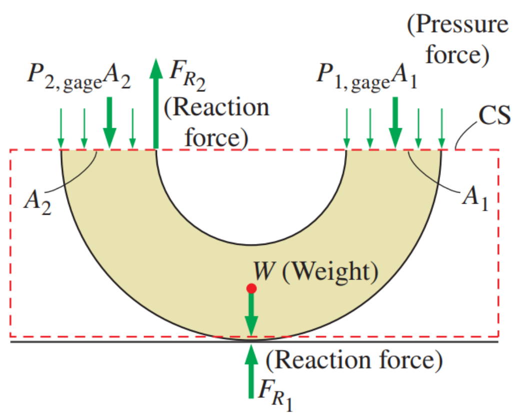
$$\frac{dB_{sys}}{dt} = \frac{d}{dt}\int_{CV} \rho b \, dV + \int_{CS} \rho b (\vec{V}_r \cdot \vec{n})\, dA$$

$$B = m\vec{V} \qquad b = \vec{V} \qquad b = \vec{V}$$

$$\frac{d(m\vec{V})_{sys}}{dt} = \frac{d}{dt}\int_{CV} \rho \vec{V} \, dV + \int_{CS} \rho \vec{V} (\vec{V}_r \cdot \vec{n})\, dA$$

The linear momentum equation is obtained by replacing *B* in the Reynolds transport theorem by the momentum $m\vec{V}$, and *b* by the momentum per unit mass $\vec{V}$.

- The momentum equation is commonly used to calculate the forces (usually on support systems or connectors) induced by the flow.



An 180° elbow supported by the ground

- In most flow systems, the sum of forces $\Sigma\vec{F}$ consists of weights, pressure forces, and reaction forces. Gage pressures are used here since atmospheric pressure cancels out on all sides of the control surface.

$$\sum \vec{F} = \frac{d}{dt}\int_{CV} \rho \vec{V} \, dV + \int_{CS} \rho \vec{V} (\vec{V} \cdot \vec{n})\, dA$$

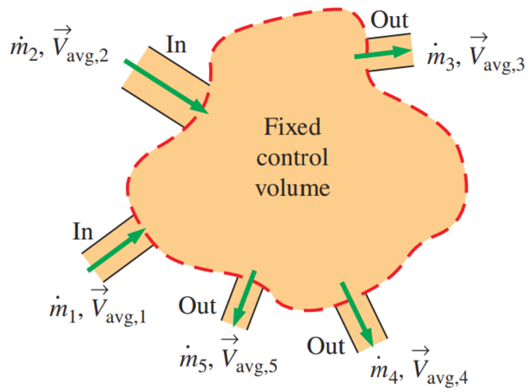
Special Cases

Steady flow: $$\sum \vec{F} = \int_{CS} \rho \vec{V} (\vec{V}_r \cdot \vec{n})\, dA$$

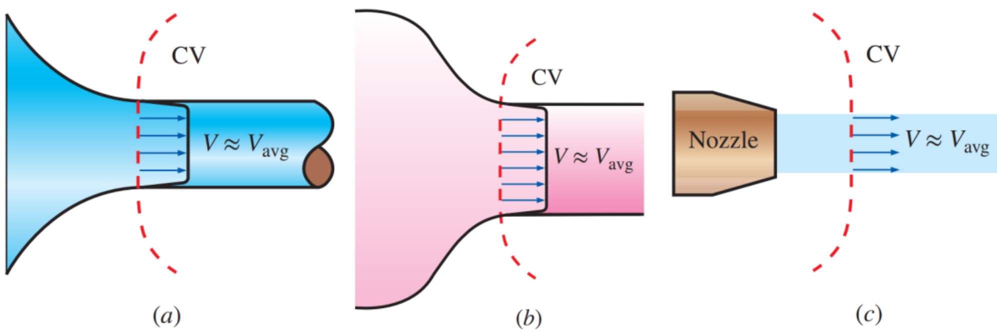
Mass flow rate across an inlet or outlet: $$\dot{m} = \int_{A_c} \rho (\vec{V} \cdot \vec{n})\, dA_c = \rho V_{avg} A_c$$

Momentum flow rate across a uniform inlet or outlet:

$$\int_{A_c} \rho \vec{V}(\vec{V}\cdot\vec{n})\, dA_c = \rho V_{avg} A_c \vec{V}_{avg} = \dot{m}\vec{V}_{avg}$$



- In a typical engineering problem, the control volume may contain multiple inlets and outlets; at each inlet or outlet we define the mass flow rate and the average velocity.



Examples of inlets or outlets in which the uniform flow approximation is reasonable:

(*a*) the well-rounded entrance to a pipe,
(*b*) the entrance to a wind tunnel test section, and
(*c*) a slice through a free water jet in air.

Topic No. 90

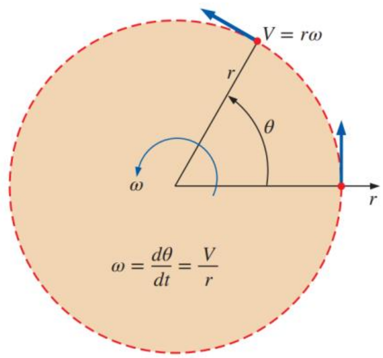
Review of Rotational Motion and Angular Momentum

- The rotational motion: A motion during which all points in the body move in circles about the axis of rotation.
- Rotational motion is described with angular quantities such as angular distance θ, angular velocity $\vec{\omega}$, and angular acceleration $\vec{\alpha}$.

$$\omega = \frac{d\theta}{dt} = \frac{d(l/r)}{dt} = \frac{1}{r}\frac{dl}{dt} = \frac{V}{r} \quad \text{and} \quad \alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2} = \frac{1}{r}\frac{dV}{dt} = \frac{a_t}{r}$$

or

$$V = r\omega \quad \text{and} \quad a_t = r\alpha$$



The relations between angular distance θ, angular velocity ω, and linear velocity *V* in a plane.

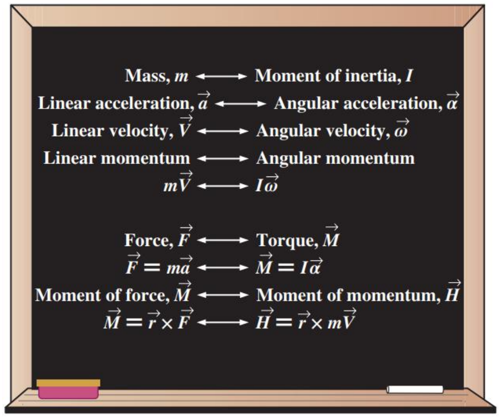
- Newton's second law requires that there must be a force acting in the tangential direction to cause angular acceleration.
- The strength of the rotating effect, called the moment or torque, is proportional to the magnitude of the force and its distance from the axis of rotation.
- The perpendicular distance from the axis of rotation to the line of action of the force is called the moment arm, and the magnitude of torque *M* acting on a point mass *m* at normal distance *r* from the axis of rotation is expressed as
- Torque

$$M = rF_t = rma_t = mr^2\alpha$$

- Magnitude of torque

$$M = \int_{\text{mass}} r^2\alpha\, \delta m = \left[\int_{\text{mass}} r^2\, \delta m\right]\alpha = I\alpha$$

- I is the moment of inertia of the body about the axis of rotation, which is a measure of the inertia of a body against rotation.
- Unlike mass, the rotational inertia of a body also depends on the distribution of the mass of the body with respect to the axis of rotation.



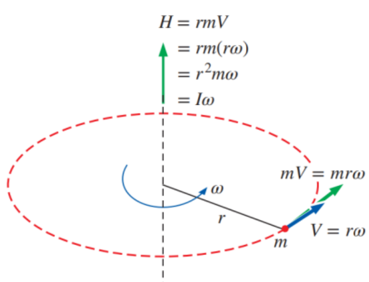
Analogy between corresponding linear and angular quantities.

Magnitude of angular momentum: $$H = \int_{\text{mass}} r^2\omega\, \delta m = \left[\int_{\text{mass}} r^2\, \delta m\right]\omega = I\omega$$

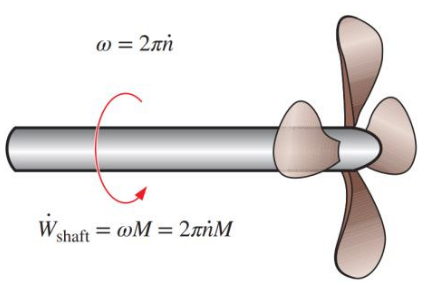
$$\vec{H} = I\vec{\omega}$$

Angular momentum equation: $$\vec{M} = I\vec{\alpha} = I\frac{d\vec{\omega}}{dt} = \frac{d(I\vec{\omega})}{dt} = \frac{d\vec{H}}{dt}$$

Angular velocity versus rpm: $$\omega = 2\pi\dot{n}\ \text{(rad/min)} = \frac{2\pi\dot{n}}{60}\quad \text{(rad/s)}$$



Angular momentum of point mass *m* rotating at angular velocity ω at distance *r* from the axis of rotation.



The relations between angular velocity, rpm, and the power transmitted through a rotating shaft.

$$\dot{W}_{\text{shaft}} = FV = Fr\omega = M\omega$$

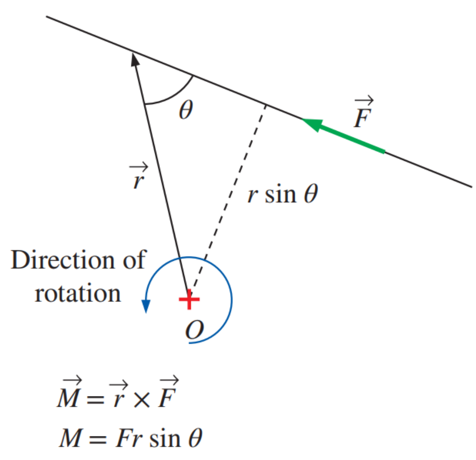
Shaft power: $$\dot{W}_{\text{shaft}} = \omega M = 2\pi \dot{n} M$$

Rotational kinetic energy: $$\text{KE}_r = \frac{1}{2} I\omega^2$$

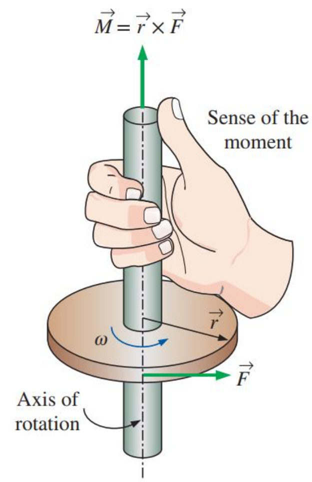
Topic No. 91

The Angular Momentum Equation-1

- Many engineering problems involve the moment of the linear momentum of flow streams, and the rotational effects caused by them.
- Such problems are best analyzed by the angular momentum equation, also called the moment of momentum equation.
- An important class of fluid devices, called turbo machines, which include centrifugal pumps, turbines, and fans, is analyzed by the angular momentum equation.
- A force whose line of action passes through point O produces zero moment about point O.



The moment of a force $\vec{F}$ about a point *O* is the vector product of the position vector $\vec{r}$ and $\vec{F}$.



The determination of the direction of the moment by the right-hand rule.

Moment of momentum: $\vec{H} = \vec{r} \times m\vec{V}$

Moment of momentum (system): $\vec{H}_{sys} = \int_{sys} (\vec{r} \times \vec{V})\rho \, dV$

Rate of change of moment of momentum: $\dfrac{d\vec{H}_{sys}}{dt} = \dfrac{d}{dt}\int_{sys} (\vec{r} \times \vec{V})\rho \, dV$

$$\sum \vec{M} = \frac{d\vec{H}_{sys}}{dt} \quad \text{where } \sum \vec{M} = \sum(\vec{r} \times \vec{F})$$

$$\frac{d\vec{H}_{sys}}{dt} = \frac{d}{dt}\int_{CV} (\vec{r} \times \vec{V})\rho \, dV + \int_{CS} (\vec{r} \times \vec{V})\rho(\vec{V}_r \cdot \vec{n}) \, dA$$

General: $$\sum \vec{M} = \frac{d}{dt}\int_{CV} (\vec{r} \times \vec{V})\rho \, dV + \int_{CS} (\vec{r} \times \vec{V})\rho(\vec{V}_r \cdot \vec{n}) \, dA$$

$$\begin{pmatrix}\text{The sum of all} \\ \text{external moments} \\ \text{acting on a CV}\end{pmatrix} = \begin{pmatrix}\text{The time rate of change} \\ \text{of the angular momentum} \\ \text{of the contents of the CV}\end{pmatrix} + \begin{pmatrix}\text{The net flow rate of} \\ \text{angular momentum} \\ \text{out of the control} \\ \text{surface by mass flow}\end{pmatrix}$$

Fixed CV: $$\sum \vec{M} = \frac{d}{dt}\int_{CV} (\vec{r} \times \vec{V})\rho \, dV + \int_{CS} (\vec{r} \times \vec{V})\rho(\vec{V} \cdot \vec{n}) \, dA$$

$$\frac{dB_{sys}}{dt} = \frac{d}{dt}\int_{CV} \rho b \, dV + \int_{CS} \rho b (\vec{V}_r \cdot \vec{n})\, dA$$

$$B = \vec{H} \qquad b = \vec{r} \times \vec{V} \qquad b = \vec{r} \times \vec{V}$$

$$\frac{d\vec{H}_{sys}}{dt} = \frac{d}{dt}\int_{CV} (\vec{r} \times \vec{V})\rho \, dV + \int_{CS} (\vec{r} \times \vec{V})\, \rho(\vec{V}_r \cdot \vec{n})\, dA$$

The angular momentum equation is obtained by replacing *B* in the Reynolds transport theorem by the angular momentum $\vec{H}$, and *b* by the angular momentum per unit mass $\vec{r} \times \vec{F}$.

Topic No. 92

The Angular Momentum Equation-2

Special Cases

A rotating lawn sprinkler is a good example of application of the angular momentum equation.

Steady flow:
$$\sum \vec{M} = \int_{CS} (\vec{r} \times \vec{V})\rho(\vec{V}_r \cdot \vec{n})\, dA$$

- An approximate form of the angular momentum equation in terms of average properties at inlets and outlets becomes

$$\sum \vec{M} \cong \frac{d}{dt}\int_{CV} (\vec{r} \times \vec{V})\rho \, dV + \sum_{out} (\vec{r} \times \dot{m}\vec{V}) - \sum_{in} (\vec{r} \times \dot{m}\vec{V})$$

Steady flow:
$$\sum \vec{M} = \sum_{out} (\vec{r} \times \dot{m}\vec{V}) - \sum_{in} (\vec{r} \times \dot{m}\vec{V})$$

- The net torque acting on the control volume during steady flow is equal to the difference between the outgoing and incoming angular momentum flow rates.
- Scalar form of angular momentum equation

$$\sum M = \sum_{\text{out}} r\dot{m}V - \sum_{\text{in}} r\dot{m}V$$

Topic No. 93

Flow with No External Moments

No external moments:
$$0 = \frac{d\vec{H}_{\text{CV}}}{dt} + \sum_{\text{out}} (\vec{r} \times \dot{m}\vec{V}) - \sum_{\text{in}} (\vec{r} \times \dot{m}\vec{V})$$

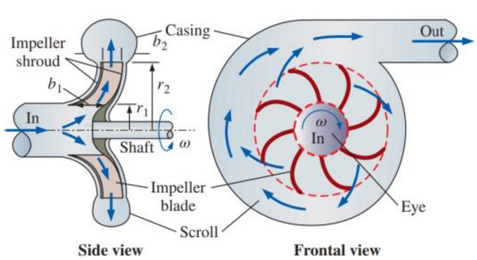
- This is an expression of the conservation of angular momentum principle, which can be stated as in the absence of external moments, the rate of change of the angular momentum of a control volume is equal to the difference between the incoming and outgoing angular momentum fluxes.
- When the moment of inertia *I* of the control volume remains constant, the first term on the right side of the above equation becomes simply moment of inertia time's angular acceleration. Therefore, the control volume in this case can be treated as a solid body, with a net torque of

$$\vec{M}_{\text{body}} = I_{\text{body}}\,\vec{\alpha} = \sum_{\text{in}} (\vec{r} \times \dot{m}\vec{V}) - \sum_{\text{out}} (\vec{r} \times \dot{m}\vec{V})$$

- This approach can be used to determine the angular acceleration of space vehicles and aircraft when a rocket is fired in a direction different than the direction of motion.

Radial-Flow Devices

- Radial-flow devices: Many rotary-flow devices such as centrifugal pumps and fans involve flow in the radial direction normal to the axis of rotation.
- Axial-flow devices are easily analyzed using the linear momentum equation.
- Radial-flow devices involve large changes in angular momentum of the fluid and are best analyzed with the help of the angular momentum equation.



Side and frontal views of a typical centrifugal pump.

- The conservation of mass equation for steady, incompressible flow

$$\dot{V}_1 = \dot{V}_2 = \dot{V} \quad \rightarrow \quad (2\pi r_1 b_1)V_{1,n} = (2\pi r_2 b_2)V_{2,n}$$

$$V_{1,n} = \frac{\dot{V}}{2\pi r_1 b_1} \quad \text{and} \quad V_{2,n} = \frac{\dot{V}}{2\pi r_2 b_2}$$

- Angular momentum equation

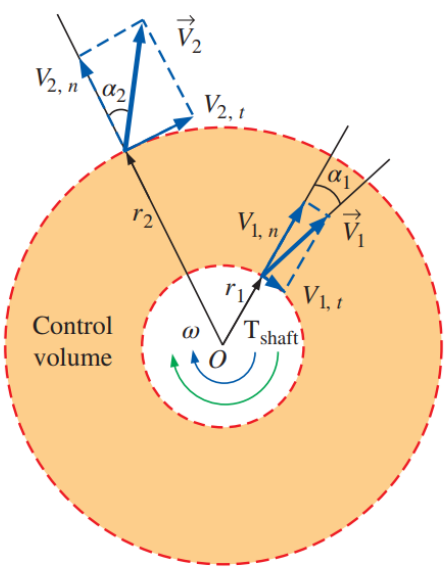
$$\sum M = \sum_{\text{out}} r\dot{m}V - \sum_{\text{in}} r\dot{m}V$$

Euler's turbine equation: $\mathrm{T}_{\text{shaft}} = \dot{m}(r_2 V_{2,t} - r_1 V_{1,t})$

$$\mathrm{T}_{\text{shaft}} = \dot{m}(r_2 V_2 \sin\alpha_2 - r_1 V_1 \sin\alpha_1)$$

$$\mathrm{T}_{\text{shaft, ideal}} = \dot{m}\omega(r_2^2 - r_1^2)$$

$$\dot{W}_{\text{shaft}} = \omega \mathrm{T}_{\text{shaft}} = 2\pi\dot{n}\mathrm{T}_{\text{shaft}}.$$



An annular control volume that encloses the impeller section of a centrifugal pump.

Topic No. 94

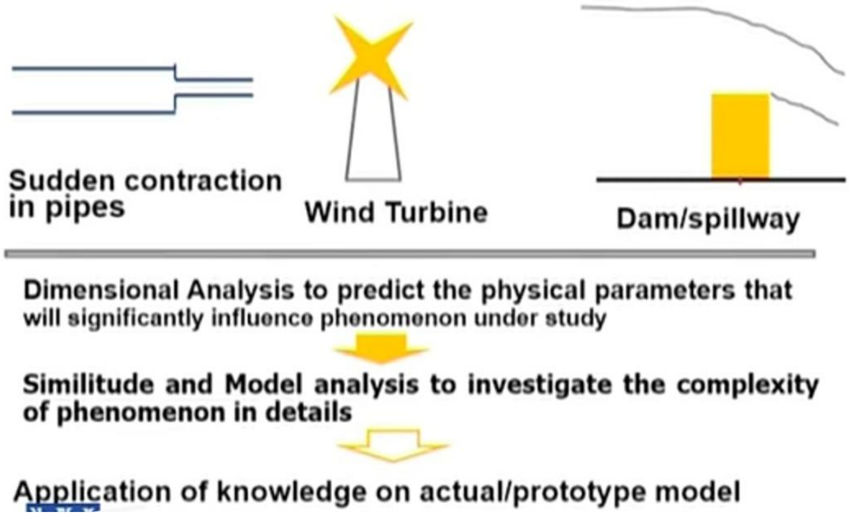
Summary

- Newton's Laws
- Choosing a Control Volume
- Forces Acting on a Control Volume
- The Linear Momentum Equation
    - Special Cases
    - Momentum-Flux Correction Factor, β
    - Steady Flow
    - Flow with No External Forces
- Review of Rotational Motion and Angular Momentum
- The Angular Momentum Equation
    - Special Cases
    - Flow with No External Moments
    - Radial-Flow Devices

Topic No. 95

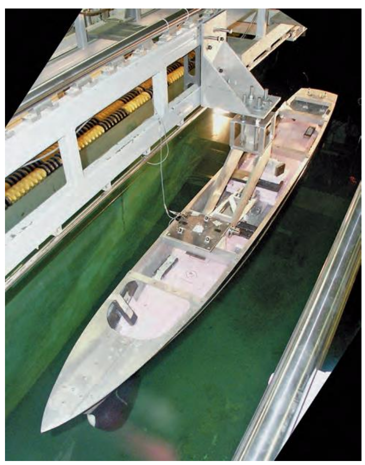
Dimensional Analysis Background

- Although many practical engineering problems involving fluid mechanics can be solved by equations and analytical procedures, yet a large number of problems rely on experimental data for their solution.
- In fact, very few problems involving real fluids can be solved by analytical analysis alone.
- In general, solution is obtained through the use of a combination of analysis and experimental data
- An obvious goal of any experiment is to make the results as widely arrears as possible
- To achieve this goal, the concept of similitude is often used so that measurements made on one system (Laboratory) can be used to describe the behavior of other systems (Outside of laboratory)
- The laboratory systems are usually thought of as models and are used to study the phenomenon of interest under carefully controlled conditions
- From these model studies, empirical formulations can be developed, or specific predications of one or more characteristics of some other similar system can be made.
- However, to do this, it is necessary to establish the relationship between the laboratory model and the "other" system
- In present topic, we will learn how to achieve this in a systematic manner.



Topic No. 96

Dimensional Analysis



A 1:46.6 scale model of an Arleigh Burke class U.S. Navy fleet destroyer being tested in the 100-m long towing tank at the University of Iowa. The model is 3.048 m long. In tests like this, the Froude number is the most important nondimensional parameter.

Objectives

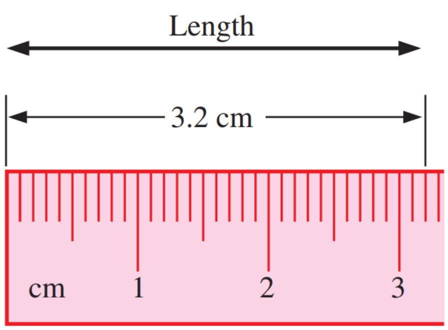
- Develop a better understanding of dimensions, units, and dimensional homo geneity of equations
- Understand the numerous benefits of dimensional analysis
- Know how to use the method of repeating variables to identify nondimensional parameters
- Understand the concept of dynamic similarity and how to apply it to experimental modeling

Topic No. 97

Dimensions and Units

- **Dimension:** A measure of a physical quantity (without numerical values).
- **Unit:** A way to assign a *number* to that dimension.
- There are seven **primary dimensions** (also called **fundamental** or **basic dimensions**) mass, length, time, temperature, electric current, amount of light, and amount of matter.
- All nonprimary dimensions can be formed by some combination of the seven primary dimensions.

$$\textit{Dimensions of force:} \quad \{\text{Force}\} = \left\{ \text{Mass}\,\frac{\text{Length}}{\text{Time}^2} \right\} = \{\text{mL/t}^2\}$$



A *dimension* is a measure of a physical quantity without numerical values, while a *unit* is a way to assign a number to the dimension. For example, length is a dimension, but centimeter is a unit.

Primary dimensions and their associated primary SI and English units			
Dimension	Symbol*	SI Unit	English Unit
Mass	m	kg (kilogram)	lbm (pound-mass)
Length	L	m (meter)	ft (foot)
Time†	t	s (second)	s (second)
Temperature	T	K (kelvin)	R (rankine)
Electric current	I	A (ampere)	A (ampere)
Amount of light	C	cd (candela)	cd (candela)
Amount of matter	N	mol (mole)	mol (mole)

Topic No. 98 & 99

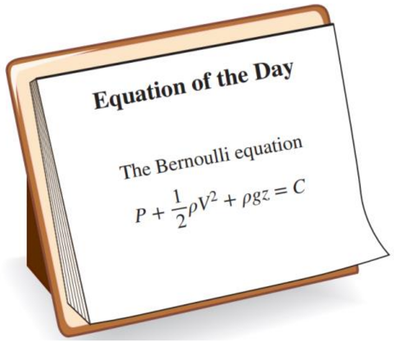
Dimensional Homogeneity

EXAMPLE 7–2 Dimensional Homogeneity of the Bernoulli Equation

Probably the most well-known (and most misused) equation in fluid mechanics is the Bernoulli equation (Fig. 7–6), discussed in Chap. 5. One standard form of the Bernoulli equation for incompressible irrotational fluid flow is

Bernoulli equation: $$P + \frac{1}{2}\rho V^2 + \rho g z = C \quad \textbf{(1)}$$

(*a*) Verify that each additive term in the Bernoulli equation has the same dimensions. (*b*) What are the dimensions of the constant C?



The Bernoulli equation is a good example of a dimensionally homogeneous equation. All additive terms, including the constant, have the *same* dimensions, namely that of pressure. In terms of primary dimensions, each term has dimensions {m/(t^{2}L)}.

SOLUTION We are to verify that the primary dimensions of each additive term in Eq. 1 are the same, and we are to determine the dimensions of constant C.

Analysis (*a*) Each term is written in terms of primary dimensions,

$$\{P\} = \{\text{Pressure}\} = \left\{\frac{\text{Force}}{\text{Area}}\right\} = \left\{\text{Mass}\,\frac{\text{Length}}{\text{Time}^2}\,\frac{1}{\text{Length}^2}\right\} = \left\{\frac{\text{m}}{\text{t}^2\text{L}}\right\}$$

$$\left\{\frac{1}{2}\rho V^2\right\} = \left\{\frac{\text{Mass}}{\text{Volume}}\left(\frac{\text{Length}}{\text{Time}}\right)^2\right\} = \left\{\frac{\text{Mass} \times \text{Length}^2}{\text{Length}^3 \times \text{Time}^2}\right\} = \left\{\frac{\text{m}}{\text{t}^2\text{L}}\right\}$$

$$\{\rho g z\} = \left\{\frac{\text{Mass}}{\text{Volume}}\,\frac{\text{Length}}{\text{Time}^2}\,\text{Length}\right\} = \left\{\frac{\text{Mass} \times \text{Length}^2}{\text{Length}^3 \times \text{Time}^2}\right\} = \left\{\frac{\text{m}}{\text{t}^2\text{L}}\right\}$$

Indeed, **all three additive terms have the same dimensions**.

(*b*) From the law of dimensional homogeneity, the constant must have the same dimensions as the other additive terms in the equation. Thus,

Primary dimensions of the Bernoulli constant: $$\{C\} = \left\{\frac{\mathbf{m}}{\mathbf{t^2L}}\right\}$$

Discussion If the dimensions of any of the terms were different from the others, it would indicate that an error was made somewhere in the analysis.

Topic No. 100

Non Dimensionalization of Equations-1

Normalized Equation:

- If the nondimensional terms in the equation are of order unity, the equation is called **normalized**.
- Each term in a nondimensional equation is dimensionless.

Nondimensional Parameters:

- In the process of nondimensionalizing an equation of motion, **nondimensional parameters** often appear—most of which are named after a notable scientist or engineer (e.g., the Reynolds number and the Froude number).
- This process is referred to by some authors as **inspectional analysis**.

Topic No. 101

Non Dimensionalization of Equations-2

The nondimensionalized Bernoulli equation

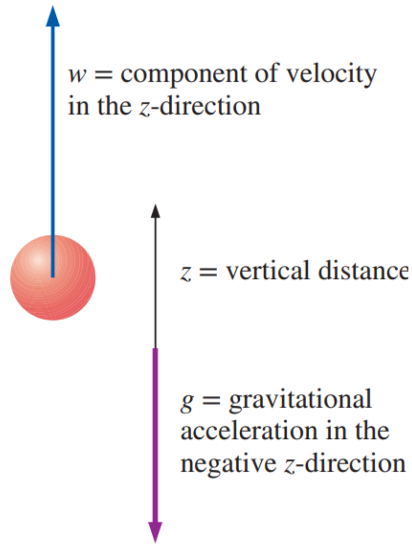
$$\underbrace{\frac{P}{P_\infty}}_{\{1\}} + \underbrace{\frac{\rho V^2}{2P_\infty}}_{\{1\}} + \underbrace{\frac{\rho g z}{P_\infty}}_{\{1\}} = \underbrace{\frac{C}{P_\infty}}_{\{1\}}$$

A nondimensionalized form of the Bernoulli equation is formed by dividing each additive term by a pressure (here we use P_∞). Each resulting term is dimensionless (dimensions of {1}).

Equation of motion: $$\frac{d^2z}{dt^2} = -g$$

Dimensional result: $$z = z_0 + w_0 t - \frac{1}{2} g t^2$$

- Object falling in a vacuum. Vertical velocity is drawn positively, so $w < 0$ for a falling object.



Topic No. 102

Non Dimensionalization of Equations-3

Dimensional variables:

- Dimensional quantities that change or vary in the problem. Examples: *z* (dimension of length) and *t* (dimension of time).

Nondimensional (or dimensionless) variables

- Quantities that change or vary in the problem, but have no dimensions. Example: Angle of rotation, measured in degrees or radians which are dimensionless units.

- **Dimensional constant**: Gravitational constant *g*, while dimensional, remains constant.

- **Parameters**: Refer to the combined set of dimensional variables, nondimensional variables, and dimensional constants in the problem.

- **Pure Constants:** The constant 1/2 and the exponent 2 in following equation. Other common examples of pure constants are π and *e*.

$$z = z_0 + w_0 t - \frac{1}{2} g t^2$$

Topic No. 103

Non Dimensionalization of Equations-4

To nondimensionalize an equation, we need to select **scaling parameters**, based on the primary dimensions contained in the original equation.

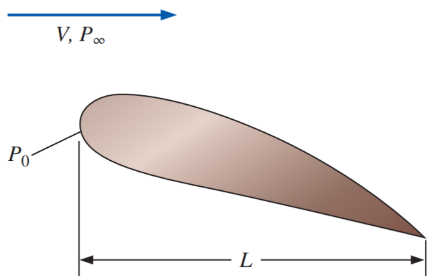
Primary dimensions of all parameters:

$$\{z\} = \{L\} \qquad \{t\} = \{t\} \qquad \{z_0\} = \{L\} \qquad \{w_0\} = \{L/t\} \qquad \{g\} = \{L/t^2\}$$

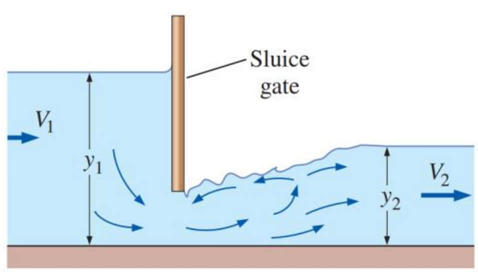
Nondimensionalized variables: $$z^* = \frac{z}{z_0} \qquad t^* = \frac{w_0 t}{z_0}$$

$$\frac{d^2 z}{dt^2} = \frac{d^2(z_0 z^*)}{d(z_0 t^*/w_0)^2} = \frac{w_0^2}{z_0}\frac{d^2 z^*}{dt^{*2}} = -g \quad \rightarrow \quad \frac{w_0^2}{g z_0}\frac{d^2 z^*}{dt^{*2}} = -1$$

Froude number: $$\mathrm{Fr} = \frac{w_0}{\sqrt{g z_0}}$$



In a typical fluid flow problem, the scaling parameters usually include a characteristic length *L*, a characteristic velocity *V*, and a reference pressure difference $P_0 - P_\infty$. Other parameters and fluid properties such as density, viscosity, and gravitational acceleration enter the problem as well.



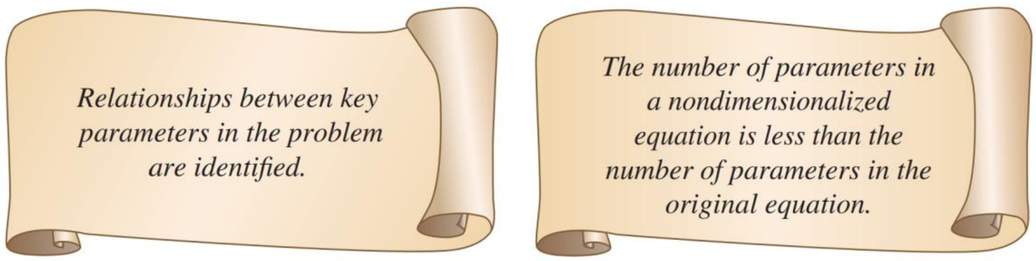
The Froude number is important in free-surface flows such as flow in open channels. Shown here is flow through a sluice gate. The Froude number upstream of the sluice gate is $Fr_1 = V_1/\sqrt{gy_1}$, and it is $Fr_2 = V_2/\sqrt{gy_2}$ downstream of the sluice gate.

Nondimensionalized equation of motion: $$\frac{d^2z^*}{dt^{*2}} = -\frac{1}{\mathrm{Fr}^2}$$

Nondimensional result: $$z^* = 1 + t^* - \frac{1}{2\mathrm{Fr}^2}t^{*2}$$

Topic No. 104

Non Dimensionalization of Equations-5



The two key advantages of nondimensionalization of an equation.

EXAMPLE 7–3 Illustration of the Advantages of Nondimensionalization

Your little brother's high school physics class conducts experiments in a large vertical pipe whose inside is kept under vacuum conditions. The students are able to remotely release a steel ball at initial height z_0 between 0 and 15 m (measured from the bottom of the pipe), and with initial vertical speed w_0 between 0 and 10 m/s. A computer coupled to a network of photosensors along the pipe enables students to plot the trajectory of the steel ball (height z plotted as a function of time t) for each test. The students are unfamiliar with dimensional analysis or nondimensionalization techniques, and therefore conduct several "brute force" experiments to determine how the trajectory is affected by initial conditions z_0 and w_0. First they hold w_0 fixed at 4 m/s and conduct experiments at five different values of z_0: 3, 6, 9, 12, and 15 m. The experimental results are shown in Fig. 7–12*a*. Next, they hold z_0 fixed at 10 m and conduct experiments at five different values of w_0: 2, 4, 6, 8, and 10 m/s. These results are shown in Fig. 7–12*b*. Later that evening, your brother shows you the data and the trajectory plots and tells you that they plan to conduct more experiments at different values of z_0 and w_0. You explain to him that by first nondimensionalizing the data, the problem can be reduced to just *one* parameter, and no further experiments are required. Prepare a nondimensional plot to prove your point and discuss.

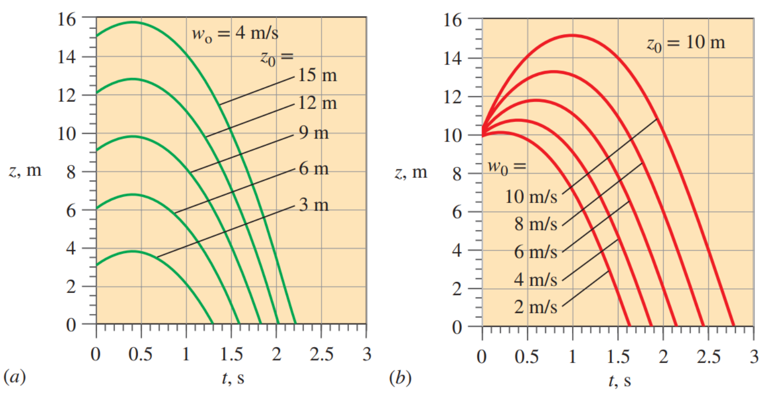
SOLUTION A nondimensional plot is to be generated from all the available trajectory data. Specifically, we are to plot z^* as a function of t^*.

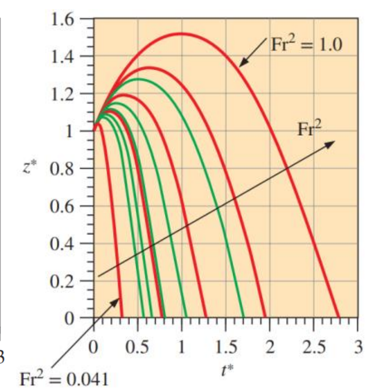
Assumptions The inside of the pipe is subjected to strong enough vacuum pressure that aerodynamic drag on the ball is negligible.

Properties The gravitational constant is 9.81 m/s^2.

Analysis Equation 7–4 is valid for this problem, as is the nondimensionalization that resulted in Eq. 7–9. As previously discussed, this problem combines three of the original dimensional parameters (g, z_0, and w_0) into *one* nondimensional parameter, the Froude number. After converting to the dimensionless variables of Eq. 7–6, the 10 trajectories of Fig. 7–12*a* and *b* are replotted in dimensionless format in Fig. 7–13. It is clear that all the trajectories are of the same family, with the Froude number as the only remaining parameter. Fr^2 varies from about 0.041 to about 1.0 in these experiments. If any more experiments are to be conducted, they should include combinations of z_0 and w_0 that produce Froude numbers outside of this range. A large number of additional experiments would be unnecessary, since all the trajectories would be of the same family as those plotted in Fig. 7–13.

Discussion At low Froude numbers, gravitational forces are much larger than inertial forces, and the ball falls to the floor in a relatively short time. At large values of Fr on the other hand, inertial forces dominate initially, and the ball rises a significant distance before falling; it takes much longer for the ball to hit the ground. The students are obviously not able to adjust the gravitational constant, but if they could, the brute force method would require many more experiments to document the effect of g. If they nondimensionalize first, however, the dimensionless trajectory plots already obtained and shown in Fig. 7–13 would be valid for *any* value of g; no further experiments would be required unless Fr were outside the range of tested values.



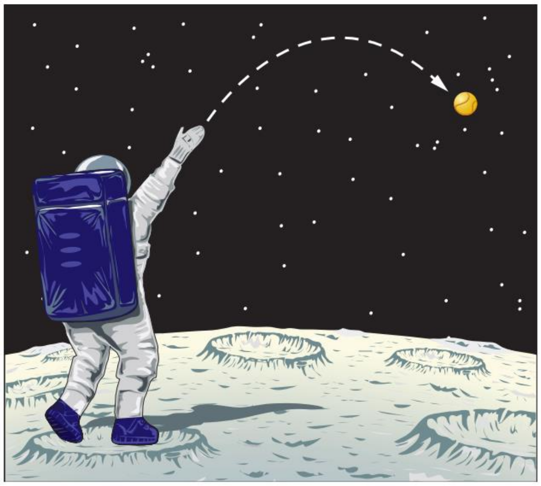


Trajectories of a steel ball falling in a vacuum: (*a*) *w*0 fixed at 4 m/s, and (*b*) *z*0 fixed at 10 m (Example 7–3).

Trajectories of a steel ball falling in a vacuum: Data of Fig. 7–12*a* and *b* are nondimensionalized and combined onto one plot.

Topic No. 105

Non Dimensionalization of Equations-6



Throwing a baseball on the moon

EXAMPLE 7–4 Extrapolation of Nondimensionalized Data

The gravitational constant at the surface of the moon is only about one-sixth of that on earth. An astronaut on the moon throws a baseball at an initial speed of 21.0 m/s at a 5° angle above the horizon and at 2.0 m above the moon's surface (Fig. 7–14). (*a*) Using the dimensionless data of Example 7–3 shown in Fig. 7–13, predict how long it takes for the baseball to fall to the ground. (*b*) Do an *exact* calculation and compare the result to that of part (*a*).

SOLUTION Experimental data obtained on earth are to be used to predict the time required for a baseball to fall to the ground on the moon.

Assumptions **1** The horizontal velocity of the baseball is irrelevant. **2** The surface of the moon is perfectly flat near the astronaut. **3** There is no aerodynamic drag on the ball since there is no atmosphere on the moon. **4** Moon gravity is one-sixth that of earth.

Properties The gravitational constant on the moon is $g_{\text{moon}} \cong 9.81/6 = 1.63\ \text{m/s}^2$.

Analysis (*a*) The Froude number is calculated based on the value of g_{moon} and the vertical component of initial speed,

$$w_0 = (21.0\ \text{m/s})\sin(5°) = 1.830\ \text{m/s}$$

from which

$$\mathrm{Fr}^2 = \frac{w_0^2}{g_{\mathrm{moon}} z_0} = \frac{(1.830 \text{ m/s})^2}{(1.63 \text{ m/s}^2)(2.0 \text{ m})} = 1.03$$

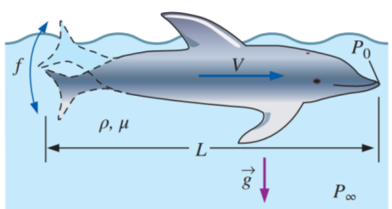
This value of Fr^2 is nearly the same as the largest value plotted in Fig. 7–13. Thus, in terms of dimensionless variables, the baseball strikes the ground at $t^* \cong 2.75$, as determined from Fig. 7–13. Converting back to dimensional variables using Eq. 7–6,

Estimated time to strike the ground: $$t = \frac{t^* z_0}{w_0} = \frac{2.75(2.0 \text{ m})}{1.830 \text{ m/s}} = \mathbf{3.01\ s}$$

(*b*) An exact calculation is obtained by setting z equal to zero in Eq. 7–5 and solving for time t (using the quadratic formula),

Topic No. 106

Non Dimensionalization of Equations-7



$\mathrm{Re} = \frac{\rho V L}{\mu}$ $\mathrm{Fr} = \frac{V}{\sqrt{gL}}$

$\mathrm{St} = \frac{fL}{V}$ $\mathrm{Eu} = \frac{P_0 - P_\infty}{\rho V^2}$

In a general unsteady fluid flow problem with a free surface, the scaling parameters include a characteristic length *L*, a characteristic velocity *V*, a characteristic frequency *f*, and a reference pressure difference $P_0 - P_\infty$. Nondimensionalization of the differential equations of fluid flow produces four dimensionless parameters: the Reynolds number, Froude number, Strouhal number, and Euler number

Topic No. 107

Dimensional Analysis and Similarity-1

- In most experiments, to save time and money, tests are performed on a geometrically scaled **model**, rather than on the full-scale **prototype**.

- In such cases, care must be taken to properly scale the results. We introduce here a powerful technique called **dimensional analysis**.

The three primary purposes of dimensional analysis are

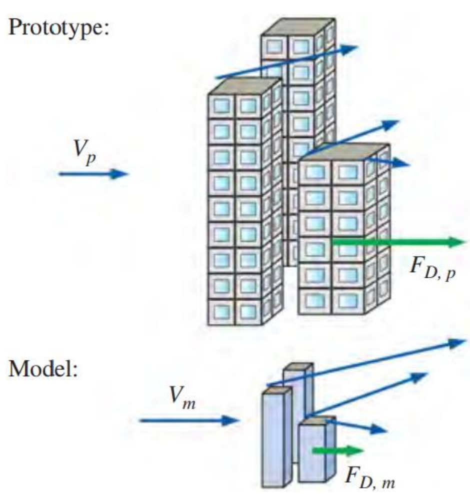
- To generate nondimensional parameters that help in the design of experiments (physical and/or numerical) and in the reporting of experimental results
- To obtain scaling laws so that prototype performance can be predicted from model performance
- To (sometimes) predict trends in the relationship between parameters

There are three necessary conditions for complete similarity between a model and a prototype.

- The first condition is **geometric similarity**, the model must be the same shape as the prototype, but may be scaled by some constant scale factor.
- The second condition is **kinematic similarity**, which means that the velocity at any point in the model flow must be proportional (by a constant scale factor) to the velocity at the corresponding point in the prototype flow.
- The third and most restrictive similarity condition is **dynamic similarity**, when all forces in the model flow scale by a constant factor to corresponding forces in the prototype flow (force-scale equivalence).

Topic No. 108

Dimensional Analysis and Similarity-2



Kinematic similarity is achieved when, at all locations, the speed in the model flow is proportional to that at corresponding locations in the prototype flow, and points in the same direction.

- In a general flow field, complete similarity between a model and prototype is achieved only when there is geometric, kinematic, and dynamic similarity.
- We let uppercase Greek letter Pi (π) denote a nondimensional parameter.
- In a general dimensional analysis problem, there is one π that we call the **dependent** π, giving it the notation π_1.
- The parameter π_1 is in general a function of several other π's, which we call **independent** π's.

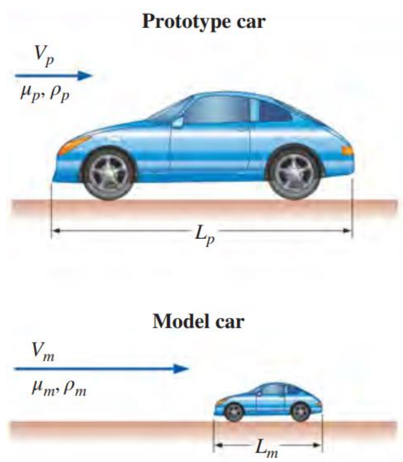
Functional relationship between Π's: $\Pi_1 = f(\Pi_2, \Pi_3, \ldots, \Pi_k)$

To ensure complete similarity, the model and prototype must be geometrically similar, and all independent π groups must match between model and prototype.

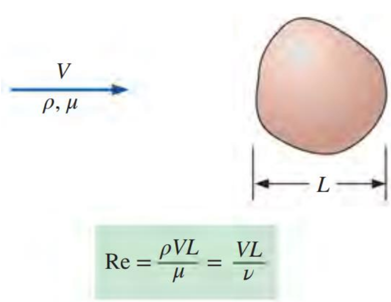
To achieve similarity

If $\Pi_{2,m} = \Pi_{2,p}$ and $\Pi_{3,m} = \Pi_{3,p} \ldots$ and $\Pi_{k,m} = \Pi_{k,p}$,

then $\Pi_{1,m} = \Pi_{1,p}$



Geometric similarity between a prototype car of length L_p and a model car of length L_m.



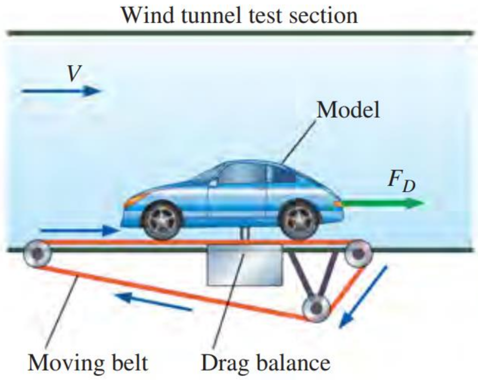
The Reynolds number Re is formed by the ratio of density, characteristic speed, and characteristic length to viscosity. Alternatively, it is the ratio of characteristic speed and length to kinematic viscosity, defined as $\nu = \mu / \rho$.

$$\Pi_1 = f(\Pi_2) \quad \text{where} \quad \Pi_1 = \frac{F_D}{\rho V^2 L^2} \quad \text{and} \quad \Pi_2 = \frac{\rho V L}{\mu}$$

The Reynolds number is the most well-known and useful dimensionless parameter in all of fluid mechanics.

Topic No. 109

Dimensional Analysis and Similarity-3



A drag balance is a device used in a wind tunnel to measure the aerodynamic drag of a body. When testing automobile models, a moving belt is often added to the floor of the wind tunnel to simulate the moving ground (from the car's frame of reference).

EXAMPLE 7–5 Similarity between Model and Prototype Cars

The aerodynamic drag of a new sports car is to be predicted at a speed of 50.0 mi/h at an air temperature of 25°C. Automotive engineers build a one-fifth scale model of the car to test in a wind tunnel. It is winter and the wind tunnel is located in an unheated building; the temperature of the wind tunnel air is only about 5°C. Determine how fast the engineers should run the wind tunnel in order to achieve similarity between the model and the prototype.

Properties For air at atmospheric pressure and at T = 25°C, ρ = 1.184 kg/m^3 and $\mu = 1.849 \times 10^{-5}$ kg/m·s. Similarly, at T = 5°C, ρ = 1.269 kg/m^3 and $\mu = 1.754 \times 10^{-5}$ kg/m·s.

Analysis Since there is only one independent Π in this problem, the similarity equation (Eq. 7–12) holds if $\Pi_{2,m} = \Pi_{2,p}$, where Π_2 is given by Eq. 7–13, and we call it the Reynolds number. Thus, we write

$$\Pi_{2,m} = \text{Re}_m = \frac{\rho_m V_m L_m}{\mu_m} = \Pi_{2,p} = \text{Re}_p = \frac{\rho_p V_p L_p}{\mu_p}$$

which we solve for the unknown wind tunnel speed for the model tests, V_m,

$$V_m = V_p\left(\frac{\mu_m}{\mu_p}\right)\left(\frac{\rho_p}{\rho_m}\right)\left(\frac{L_p}{L_m}\right)$$

$$= (50.0\ \text{mi/h})\left(\frac{1.754 \times 10^{-5}\ \text{kg/m}\cdot\text{s}}{1.849 \times 10^{-5}\ \text{kg/m}\cdot\text{s}}\right)\left(\frac{1.184\ \text{kg/m}^3}{1.269\ \text{kg/m}^3}\right)(5) = \mathbf{221\ mi/h}$$

Thus, to ensure similarity, the wind tunnel should be run at 221 mi/h (to three significant digits). Note that we were never given the actual length of either car, but the ratio of L_p to L_m is known because the prototype is five times larger than the scale model. When the dimensional parameters are rearranged as nondimensional ratios (as done here), the unit system is irrelevant. Since the units in each numerator cancel those in each denominator, no unit conversions are necessary.

Discussion This speed is quite high (about 100 m/s), and the wind tunnel may not be able to run at that speed. Furthermore, the incompressible approximation may come into question at this high speed (we discuss this in more detail in Example 7–8).

Topic No. 110

Dimensional Analysis and Similarity-4

- Nature of Dimensional Analysis
- Buckingham Pi Theorem
- Significant Dimensionless Groups in Fluid Mechanics
- Flow Similarity and Model Studies
- Understand dimensions, units, and dimensional homogeneity
- Understand benefits of dimensional analysis
- Know how to use the method of repeating variables
- Understand the concept | of similarity and how to apply it to experimental modeling
- All non-primary dimensions can be formed by a combination of the 7 primary dimensions
- Examples
  {Velocity} m/sec = {Length/Time} = {L/t}
  {Force} N = {Mass.Length/Time} = {mL/t^2}
- Every additive term in an equation must have the same dimensions
- Example: Bernoulli equation

$$p + \frac{1}{2}\rho V^2 + \rho g\, z = C$$

Nondimensionalization of Equations

$$p + \frac{1}{2}\rho V^2 + \rho g\, z = C$$

{p} = {force/area}={mass × length/time × 1/length^2} = {m/(t^{2}L)}
{1/2pV^2} = {mass/length^3 × (length/time)^2} = {m/(t^{2}L)}
{ρgz} = {mass/length^3 × length/time^2 × length} = {m/(t^{2}L)}

For B equation

{p} = {m/(t^{2}L)}
{ρ} = {m/L^3}
{V} = {L/t}
{g} = {L/t^2}
{z} = {L}

Next, we need to select Scaling Parameters. For this example, select L,U_0,ρ_0

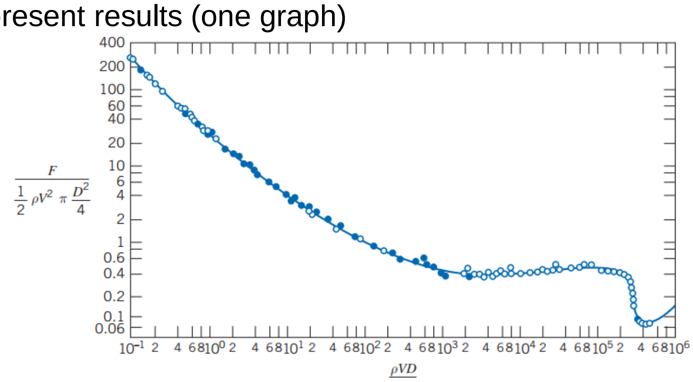
Nature of Dimensional Analysis

- Example: Drag on a Sphere
  $$F = f(D, V,\ \rho,\ \mu)$$
- Drag depends on FOUR parameters: sphere size (D); speed (V); fluid density (ρ); fluid viscosity (μ)
- Difficult to know how to set up experiments to determine dependencies
- Difficult to know how to. Present results (four graphs?)

Example: Drag on a Sphere

$$\frac{F}{\rho V^2 D^2} = f\left(\frac{\rho VD}{\mu}\right)$$

- Only one dependent and one independent | variable
- Easy to set up experiments to determine dependency
- Easy to present results (one graph)



Experimentally derived relation between the nondimensional parameters [3].

Topic No. 111

Dimensional Analysis and Similarity-5

Buckingham Pi Theorem

Step 1:

- List all the dimensional parameters involved
- Let n be the number of parameters
- Example: For drag on a sphere, F, V, D, ρ, μ & n = 5

Step 2:

- Select a set of fundamental (primary) dimensions.
- For Example M(kg), L(m), t(sec).
- Example: For drag on a sphere choose MLt

Step 3:

- List the dimensions of all parameters
- Let r be the number of primary dimensions.
- Example: For drag on a sphere r = 3

F	V	D	ρ	μ
$\frac{ML}{t^2}$	$\frac{L}{t}$	L	$\frac{M}{L^3}$	$\frac{M}{Lt}$

Step 4:

- Select a set of r dimensional parameters that includes all the primary dimensions.
- Example: For drag on a sphere (m = r = 3) select ρ, V, D

Step 5:

- Set up dimensionless groups π^s
- There will be n - m equations
- Example: For drag on a sphere

$$\Pi_1 = \rho^a V^b D^c F \quad \text{and} \quad \left(\frac{M}{L^3}\right)^a \left(\frac{L}{t}\right)^b (L)^c \left(\frac{ML}{t^2}\right) = M^0 L^0 t^0$$

$$\Pi_1 = \frac{F}{\rho V^2 D^2}$$

Step 6:

- Check to see that each group obtained is dimensionless
- Example: For drag on a sphere

$$[\Pi_1] = \left[\frac{F}{\rho V^2 D^2}\right] \quad \text{and} \quad F\frac{L^4}{Ft^2}\left(\frac{t}{L}\right)^2\frac{1}{L^2} = 1$$

$$\left(\frac{M}{L^3}\right)^a\left(\frac{L}{t}\right)^b(L)^c\left(\frac{M}{Lt^2}\right) = M^0L^0t^0 \qquad \Pi_1 = \frac{F}{\rho V^2 D^2}$$

Significant Dimensionless Groups in Fluid Mechanics

Reynolds Number

$$Re = \frac{\rho VL}{\mu} = \frac{VL}{\nu}$$

Mach Number

$$M = \frac{V}{c}$$

Froude Number

$$Fr = \frac{V}{\sqrt{gL}}$$

Weber Number

$$We = \frac{\rho V^2 L}{\sigma}$$

Euler Number

$$Eu = \frac{\Delta p}{\frac{1}{2}\rho V^2}$$

Cavitation Number

$$Ca = \frac{p - p_v}{\frac{1}{2}\rho V^2}$$

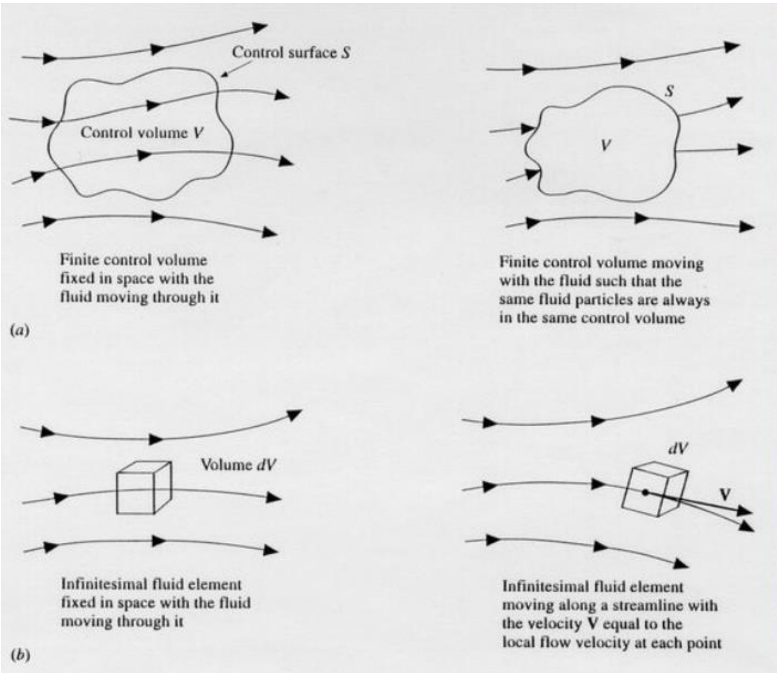
Topic No. 112

Differential Analysis of Fluid Flow; Introduction

- Continuity Equation
- Momentum Equation
- Euler's Equations
- Navier Stokes Equations
- Applications of the above equations
- Special cases
- Equations of Fluid Dynamics, Physical Meaning of the terms.
- Equations are based on the following physical principles:
- Mass is conserved
- Newton's Second Law: $\bar{F} = m\bar{a}$
- Control Volume Analysis
- The governing equations can be obtained in the integral form by choosing a control volume (CV) in the flow field and applying the principles of the conservation of mass, momentum and energy to the CV.

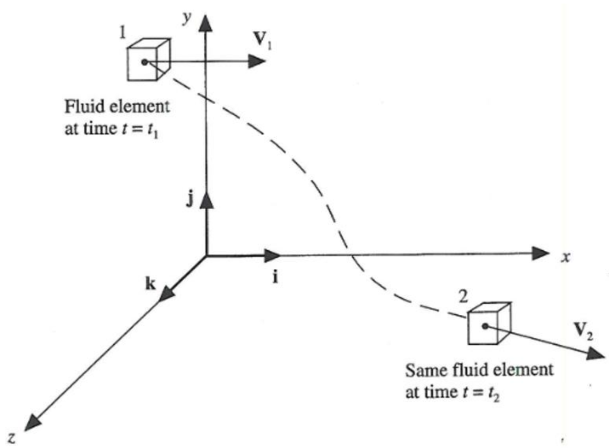
Topic No. 113

Differential Analysis of Fluid Flow; Preliminaries-1



Models of flow: (a) Finite control volume approach
(b) Infinitesimal fluid element approach

- Consider a differential volume element dv in the flow field. dv is small enough to be considered infinitesimal but large enough to contain a large number of molecules for continuum approach to be valid.
- dv may be: fixed in space with fluid flowing in and out of its surface or, moving so as to contain the same fluid particles all the time, In this case the boundaries may distort and the volume may change.



Substantial derivative (time rate of change following a moving fluid element)

Topic No. 114

Differential Analysis of Fluid Flow; Preliminaries-2

- The velocity vector can be written in terms of its Cartesian components as:

$$\vec{V} = u\vec{i} + v\vec{j} + w\vec{k}$$

where the *x, y*, and *z* components of velocity are given respectively by

$$u = u(x, y, z, t)$$
$$v = v(x, y, z, t)$$
$$w = w(x, y, z, t)$$

At time t_1:

$$\rho_1 = \rho(x_1, y_1, z_1, t_1)$$

At time t_2:

$$\rho_2 = \rho(x_2, y_2, z_2, t_2)$$

Using Taylor's series

$$\rho_2 = \rho_1 + \left(\frac{\partial \rho}{\partial x}\right)_1 (x_2 - x_1) + \left(\frac{\partial \rho}{\partial y}\right)_1 (y_2 - y_1) + \left(\frac{\partial \rho}{\partial z}\right)_1 (z_2 - z_1)$$
$$+ \left(\frac{\partial \rho}{\partial t}\right)_1 (t_2 - t_1) + (\text{higher order terms})$$

Dividing by (t_2 –t_2), and ignoring higher order terms, we obtain

$$\frac{\rho_2 - \rho_1}{t_2 - t_1} = \left(\frac{\partial \rho}{\partial x}\right)_1 \left(\frac{x_2 - x_1}{t_2 - t_1}\right) + \left(\frac{\partial \rho}{\partial y}\right)_1 \left(\frac{y_2 - y_1}{t_2 - t_1}\right)$$
$$+ \left(\frac{\partial \rho}{\partial z}\right)_1 \left(\frac{z_2 - z_1}{t_2 - t_1}\right) + \left(\frac{\partial \rho}{\partial t}\right)_1$$

We can also write

$$\lim_{t_2 \to t_1} \left(\frac{\rho_2 - \rho_1}{t_2 - t_1}\right) \equiv \frac{D\rho}{Dt}$$

$$\lim_{t_2 \to t_1} \left(\frac{x_2 - x_1}{t_2 - t_1}\right) \equiv u$$

$$\lim_{t_2 \to t_1} \left(\frac{y_2 - y_1}{t_2 - t_1}\right) \equiv v$$

$$\lim_{t_2 \to t_1} \left(\frac{z_2 - z_1}{t_2 - t_1}\right) \equiv w$$

$$\frac{D\rho}{Dt} = u\frac{\partial \rho}{\partial x} + v\frac{\partial \rho}{\partial y} + w\frac{\partial \rho}{\partial z} + \frac{\partial \rho}{\partial t}$$

where the operator $\frac{D}{Dt}$ can now be seen to be defined in the following manner.

$$\frac{D}{Dt} \equiv \frac{\partial}{\partial t} + u\frac{\partial}{\partial x} + v\frac{\partial}{\partial y} + w\frac{\partial}{\partial z}$$

Topic No. 115

Differential Analysis of Fluid Flow; Preliminaries-3

The $\vec{\nabla}$ operator in vector calculus is defined as

$$\nabla \equiv \vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}$$

which can be used to write the total derivative as

$$\frac{D}{Dt} \equiv \frac{\partial}{\partial t} + \left(\vec{V} \cdot \nabla\right)$$

Example: derivative of temperature, T

$$\frac{DT}{Dt} \equiv \underbrace{\frac{\partial T}{\partial t}}_{\substack{\text{local} \\ \text{derivative}}} + \underbrace{(\vec{V} \cdot \nabla)}_{\substack{\text{convective} \\ \text{derivative}}} T \equiv \frac{\partial T}{\partial t} + u\frac{\partial T}{\partial x} + v\frac{\partial T}{\partial y} + w\frac{\partial T}{\partial z}$$

A simpler way of writing the total derivative is as follows:

$$\mathrm{d}\rho = \frac{\partial \rho}{\partial x}\mathrm{d}x + \frac{\partial \rho}{\partial y}\mathrm{d}y + \frac{\partial \rho}{\partial z}\mathrm{d}z + \frac{\partial \rho}{\partial t}\mathrm{d}t$$

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} = \frac{\partial \rho}{\partial t} + \frac{\partial \rho}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial \rho}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t} + \frac{\partial \rho}{\partial z}\frac{\mathrm{d}z}{\mathrm{d}t}$$

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} = \frac{\partial \rho}{\partial t} + u\frac{\partial \rho}{\partial x} + v\frac{\partial \rho}{\partial y} + w\frac{\partial \rho}{\partial z}$$

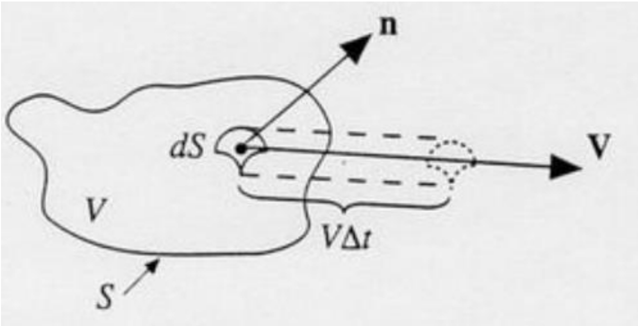
The above equation shows that $\frac{d\rho}{dt}$ and $\frac{D\rho}{Dt}$ have the same meaning, and the latter form is used simply to emphasize the physical meaning that it consists of the local derivative and the convective derivatives.

Divergence of Velocity (What does it mean?)

Consider a control volume moving with the fluid.
Its mass is fixed with respect to time.
Its volume and surface change with time as it moves from one location to another.

Topic No. 116

Differential Analysis of Fluid Flow; Preliminaries-4



Moving control volume used for the physical interpretation of the divergence of velocity.

The volume swept by the elemental area dS during time interval Dt can be written as

$$\Delta \mathcal{V} = \left[(\vec{V}\Delta t)\cdot\vec{n}\right]\mathrm{d}S = (\vec{V}\Delta t)\cdot\vec{\mathrm{d}}S$$

Note that, depending on the orientation of the surface element, Dv could be positive or negative. Dividing by Dt and letting $\Delta t \to 0$ gives the following expression.

$$\frac{D\mathcal{V}}{Dt} = \frac{1}{\Delta t}\oiint_S (\vec{V}\cdot\Delta t)\cdot\mathrm{d}\vec{S} = \oiint_S \vec{V}\cdot\mathrm{d}\vec{S}$$

The LHS term is written as a total time derivative because the fluid element is moving with the flow and it would undergo both the local acceleration and the convective acceleration.

The divergence theorem from vector calculus can now be used to transform the surface integral into a volume integral.

$$\frac{D\mathcal{V}}{Dt} = \oiiint_V (\nabla\cdot\vec{V})\mathrm{d}\mathcal{V}$$

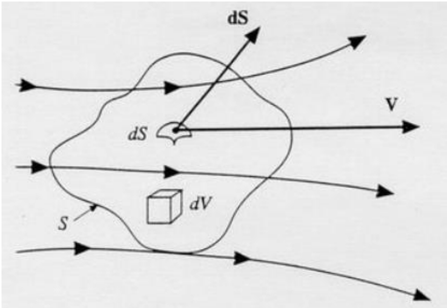
If we now shrink the moving control volume to an infinitesimal volume, $\delta\mathcal{V}$ the above equation becomes

$$\frac{D(\delta\mathcal{V})}{Dt} = \oiiint_{\delta\mathcal{V}} (\nabla\cdot\vec{V})\mathrm{d}\mathcal{V}$$

When $\delta\mathscr{V} \rightarrow 0$ the volume integral can be replaced by $\nabla \cdot \vec{V}\ \delta\mathscr{V}$ on the RHS to get the following.

$$\nabla \cdot \vec{V} = \frac{1}{\delta\mathscr{V}} \frac{D(\delta\mathscr{V})}{Dt}$$

The divergence of $\vec{V}$ is the rate of the change of volume per unit volume.



Finite control volume fixed in space

- Consider the CV fixed in space. Unlike the earlier case the shape and size of the CV are the same at all times. The conservation of mass can be stated as:
- Net rate of outflow of mass from CV through surface S = time rate of decrease of mass inside the CV

The net outflow of mass from the CV can be written as

$$\rho V_{\mathrm{n}} \mathrm{d}S = \rho \vec{V} \cdot \vec{\mathrm{d}}S$$

Note that $d\vec{S}$ by convention is always pointing outward. Therefore $\vec{V} \cdot d\vec{S}$ can be (+) or (-) depending on the directions of the velocity and the surface element.

Topic No. 117

Differential Analysis of Fluid Flow; Continuity Equation

Total mass inside CV

$$\oiiint_{\mathscr{V}} \rho \, \mathrm{d}\mathscr{V}$$

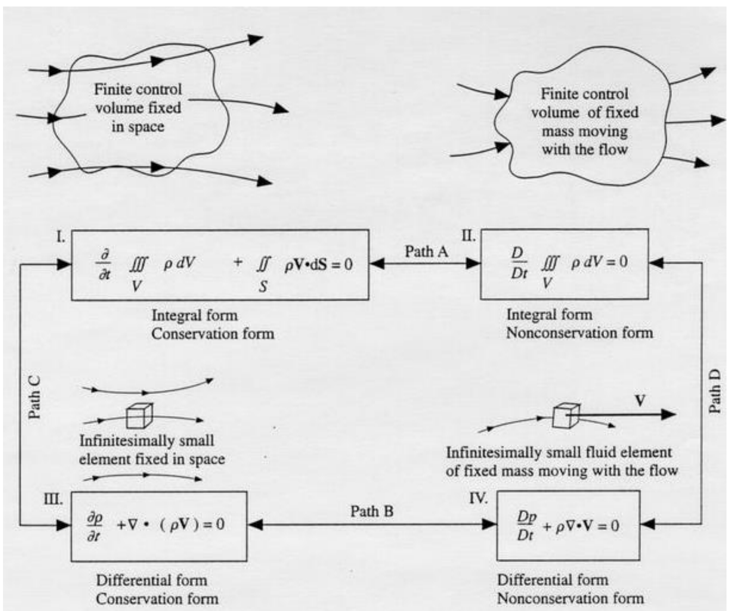
Time rate of increases of mass inside CV (correct this equation)

$$-\frac{\partial}{\partial t}\iiint_{\mathcal{V}} \rho \, \mathrm{d}\mathcal{V}$$

$$\frac{\partial}{\partial t}\iiint_{\mathcal{V}} \rho \, \mathrm{d}\mathcal{V} + \iint_{S} \rho \vec{V} \cdot \vec{\mathrm{d}S} = 0$$

Conservation of mass can now be used to write the following equation [There are other ways of obtaining the same equation as well.]

$$\frac{D}{Dt}\oiiint_{V} \rho d V = 0$$



The different forms of the continuity equation, their relationship to the different models if the flow, and the schematic emphasis that all four equations are essentially the same they can each be obtained from the other.

Topic No. 118

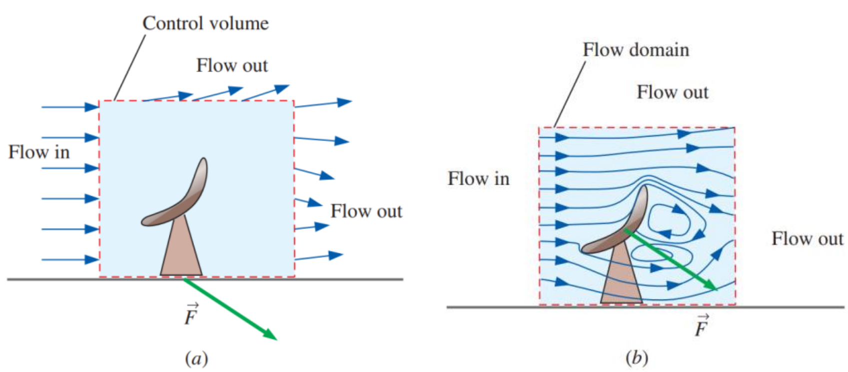
Differential Analysis of Fluid Flow; Objectives

- Now we derive the fundamental differential equations of fluid motion show how to solve them analytically for some **simple** flows. More complicated flows, such as the air flow induced by a tornado, cannot be solved exactly.
- Understand how the differential equation of conservation of mass and the differential linear momentum equation are derived and applied.

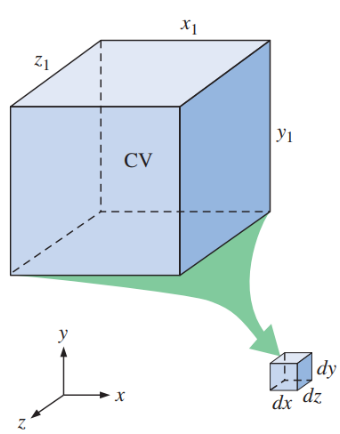
- Calculate the stream function and pressure field, and plot streamlines for a known velocity field.
- Obtain analytical solutions of the equations of motion for simple flow fields.
- The control volume technique is useful when we are interested in the overall features of a flow, such as mass flow rate into and out of the control volume or net forces applied to bodies.
- Differential analysis, on the other hand, involves application of differential equations of fluid motion to any and every point in the flow field over a region called the flow domain.
- Boundary conditions for the variables must be specified at all boundaries of the flow domain, including inlets, outlets, and walls.
- If the flow is unsteady, we must march our solution along in time as the flow field changes

Topic No. 119

Derivation of Continuity Equation



(a) In control volume analysis, the interior of the control volume is treated like a black box, but (b) in differential analysis, all the details of the flow are solved at every point within the flow domain.



To derive a differential conservation equation, we imagine shrinking a control volume to infinitesimal size.

Conservation of Mass

Conservation of mass for a CV:

$$0 = \int_{CV} \frac{\partial \rho}{\partial t}\, dV + \int_{CS} \rho \vec{V} \cdot \vec{n}\, dA$$

$$\int_{CV} \frac{\partial \rho}{\partial t}\, dV = \sum_{in} \dot{m} - \sum_{out} \dot{m}$$

The net rate of change of mass within the control volume is equal to the rate at which mass flows into the control volume minus the rate at which mass flows out of the control volume.

THE CONTINUITY EQUATION: Derivation Using the Divergence Theorem

- The quickest and most straightforward way to derive the differential form of conservation of mass is to apply the divergence theorem (Gauss's theorem).

Divergence theorem: $$\int_V \vec{\nabla} \cdot \vec{G}\, dV = \oint_A \vec{G} \cdot \vec{n}\, dA$$

$$0 = \int_{CV} \frac{\partial \rho}{\partial t}\, dV + \int_{CV} \vec{\nabla} \cdot (\rho \vec{V})\, dV$$

$$\int_{CV} \left[\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot (\rho \vec{V}) \right] dV = 0$$

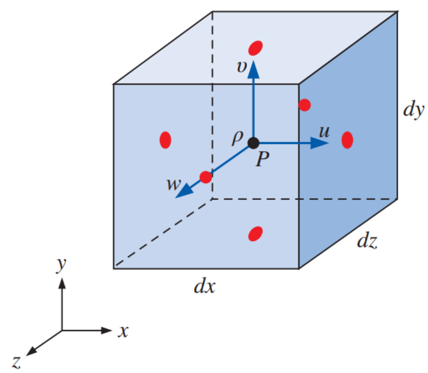
Continuity equation: $$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot (\rho \vec{V}) = 0$$

- This equation is the compressible form of the continuity equation since we have not assumed incompressible flow. It is valid at any point in the flow domain.

Topic No. 120

THE CONTINUITY EQUATION: Derivation Using an Infinitesimal Control Volume

- A small box-shaped control volume centered at point *P* is used for derivation of the differential equation for conservation of mass in Cartesian coordinates; the red dots indicate the center of each face.



- At locations away from the center of the box, we use a **Taylor series expansion** about the center of the box (point *P*).

$$(\rho u)_{\text{center of right face}} = \rho u + \frac{\partial(\rho u)}{\partial x}\frac{dx}{2} + \frac{1}{2!}\frac{\partial^2(\rho u)}{\partial x^2}\left(\frac{dx}{2}\right)^2 + \cdots$$

Center of right face: $$(\rho u)_{\text{center of right face}} \cong \rho u + \frac{\partial(\rho u)}{\partial x}\frac{dx}{2}$$

Center of left face: $$(\rho u)_{\text{center of left face}} \cong \rho u - \frac{\partial(\rho u)}{\partial x}\frac{dx}{2}$$

Center of front face: $$(\rho w)_{\text{center of front face}} \cong \rho w + \frac{\partial(\rho w)}{\partial z}\frac{dz}{2}$$

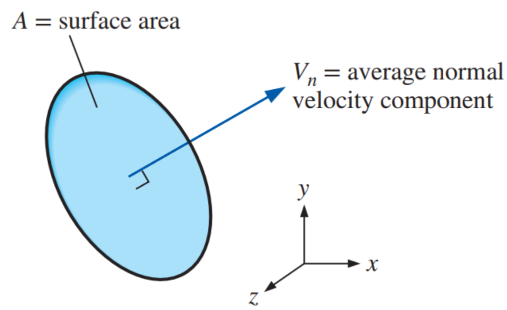
Center of rear face: $$(\rho w)_{\text{center of rear face}} \cong \rho w - \frac{\partial(\rho w)}{\partial z}\frac{dz}{2}$$

Center of top face: $$(\rho v)_{\text{center of top face}} \cong \rho v + \frac{\partial(\rho v)}{\partial y}\frac{dy}{2}$$

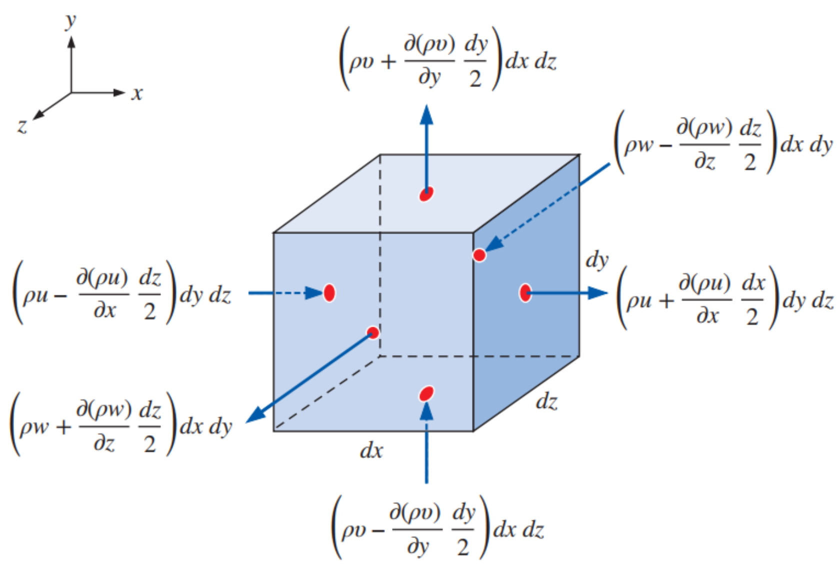
Center of bottom face: $$(\rho v)_{\text{center of bottom face}} \cong \rho v - \frac{\partial(\rho v)}{\partial y}\frac{dy}{2}$$

Topic No. 121

THE CONTINUITY EQUATION: Derivation Using an Infinitesimal Control Volume



The mass flow rate through a surface is equal to $\rho V_n A$.



The inflow or outflow of mass through each face of the differential control volume; the red dots indicate the center of each face.

Rate of change of mass within CV:

$$\int_{\mathrm{CV}} \frac{\partial \rho}{\partial t}\, dV \cong \frac{\partial \rho}{\partial t}\, dx\, dy\, dz$$

Net mass flow rate into CV:

$$\sum_{\text{in}} \dot{m} \cong \underbrace{\left(\rho u - \frac{\partial(\rho u)}{\partial x}\frac{dx}{2}\right) dy\, dz}_{\text{left face}} + \underbrace{\left(\rho v - \frac{\partial(\rho v)}{\partial y}\frac{dy}{2}\right) dx\, dz}_{\text{bottom face}} + \underbrace{\left(\rho w - \frac{\partial(\rho w)}{\partial z}\frac{dz}{2}\right) dx\, dy}_{\text{rear face}}$$

Similarly, the right, top, and front faces contribute to mass outflow

Net mass flow rate out of CV:

$$\sum_{\text{out}} \dot{m} \cong \underbrace{\left(\rho u + \frac{\partial(\rho u)}{\partial x}\frac{dx}{2}\right) dy\, dz}_{\text{right face}} + \underbrace{\left(\rho v + \frac{\partial(\rho v)}{\partial y}\frac{dy}{2}\right) dx\, dz}_{\text{top face}} + \underbrace{\left(\rho w + \frac{\partial(\rho w)}{\partial z}\frac{dz}{2}\right) dx\, dy}_{\text{front face}}$$

We substitute the equation

$$\int_{\text{CV}} \frac{\partial \rho}{\partial t}\, dV \cong \frac{\partial \rho}{\partial t}\, dx\, dy\, dz$$

and these two equations for mass flow rate into the basic equation

$$\int_{\text{CV}} \frac{\partial \rho}{\partial t}\, dV = \sum_{\text{in}} \dot{m} - \sum_{\text{out}} \dot{m}$$

Many of the terms cancel each other out; after combining and simplifying the remaining terms, we are left with

$$\frac{\partial \rho}{\partial t}\, dx\, dy\, dz = -\frac{\partial(\rho u)}{\partial x}\, dx\, dy\, dz - \frac{\partial(\rho v)}{\partial y}\, dx\, dy\, dz - \frac{\partial(\rho w)}{\partial z}\, dx\, dy\, dz$$

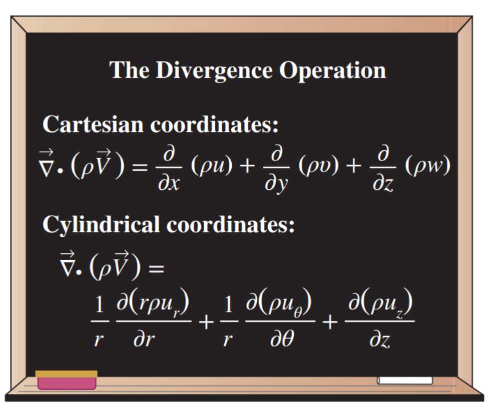
The volume of the box, *dx dy dz*, appears in each term and can be eliminated. After rearrangement we end up with the following differential equation for conservation of mass in Cartesian coordinates:

Continuity equation in Cartesian coordinates:

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0$$

Topic No. 122

THE CONTINUITY EQUATION: Derivation Using an Infinitesimal Control Volume



The divergence operation in Cartesian and cylindrical coordinates.

We expand equation

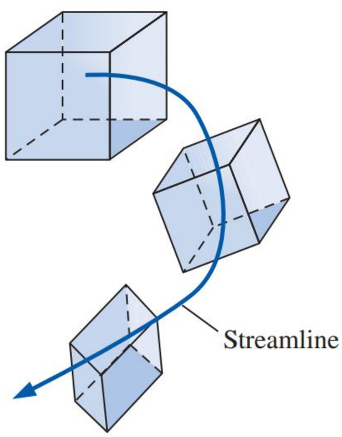
$$\frac{\partial \rho}{\partial t} + \vec{\nabla}\cdot(\rho\vec{V}) = 0$$

by using the product rule on the divergence term,

$$\frac{\partial \rho}{\partial t} + \vec{\nabla}\cdot(\rho\vec{V}) = \underbrace{\frac{\partial \rho}{\partial t} + \vec{V}\cdot\vec{\nabla}\rho}_{\text{Material derivative of }\rho} + \rho\vec{\nabla}\cdot\vec{V} = 0$$

Alternative form of the continuity equation:

$$\frac{1}{\rho}\frac{D\rho}{Dt} + \vec{\nabla}\cdot\vec{V} = 0$$



As a material element moves through a flow field, its density changes according to above equation.

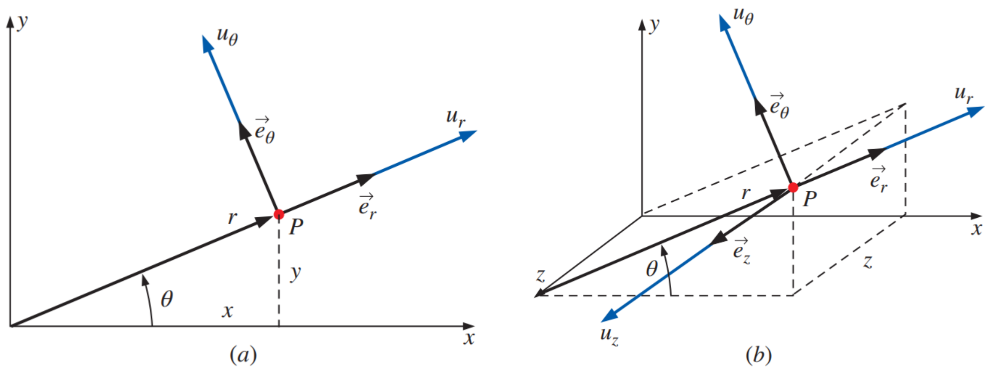
Above equation also shows that as we follow a fluid element through the flow field (we call this a **material element**), its density changes as $\vec{\nabla}\cdot\vec{V}$ changes (Figure).

- On the other hand, if changes in the density of the material element are negligibly small compared to the magnitude of the velocity gradients in $\vec{\nabla}\cdot\vec{V} \cong 0$ as the element moves around, $\rho^{-1}D\rho/Dt \cong 0$ and the flow is approximated as **incompressible**.

Topic No. 123

THE CONTINUITY EQUATION: Continuity Equation in Cylindrical Coordinates

- Many problems in fluid mechanics are more conveniently solved in **cylindrical coordinates** (r, θ, z) (often called **cylindrical polar coordinates**), rather than in Cartesian coordinates. For simplicity, we introduce cylindrical coordinates in two dimensions first {Fig. (a)}. By convention, r is the radial distance from the origin to some point (P), and θ is the angle measured from the x-axis (θ is always defined as mathematically positive in the counter clockwise direction). Velocity components, u_r and u_θ, and unit vectors, $\vec{e}_r$ and $\vec{e}_\theta$, are also shown in Fig. (a). In three dimensions, imagine sliding everything in Fig. (a) out of the page along the z-axis (normal to the xy-plane) by some distance z. We have attempted to draw this in Fig. (b). In three dimensions, we have a third velocity component, u_z, and a third unit vector, $\vec{e}_z$, also sketched in Fig. (b)



Velocity components and unit vectors in cylindrical coordinates:
(*a*) Two-dimensional flow in the *xy*- or $r\theta$-plane,
(*b*) Three-dimensional flow.

Coordinate transformations:

$$r = \sqrt{x^2 + y^2} \qquad x = r\cos\theta \qquad y = r\sin\theta \qquad \theta = \tan^{-1}\frac{y}{x}$$

Coordinate z is the same in cylindrical and Cartesian coordinates.

Continuity equation in cylindrical coordinates:

$$\frac{\partial \rho}{\partial t} + \frac{1}{r}\frac{\partial(r\rho u_r)}{\partial r} + \frac{1}{r}\frac{\partial(\rho u_\theta)}{\partial \theta} + \frac{\partial(\rho u_z)}{\partial z} = 0$$

Topic No. 124

THE CONTINUITY EQUATION: Special Cases of the Continuity Equation

Special Case 1: Steady Compressible Flow

Steady continuity equation: $\vec{\nabla}\cdot(\rho\vec{V}) = 0$

In Cartesian coordinates, this reduces to

$$\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0$$

In cylindrical coordinates, it reduces to

$$\frac{1}{r}\frac{\partial(r\rho u_r)}{\partial r} + \frac{1}{r}\frac{\partial(\rho u_\theta)}{\partial \theta} + \frac{\partial(\rho u_z)}{\partial z} = 0$$

Special Case 2: Incompressible Flow

Incompressible continuity equation: $\vec{\nabla}\cdot\vec{V} = 0$

Since $\partial\rho / \partial t \cong 0$

Incompressible continuity equation in Cartesian coordinates:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

Incompressible continuity equation in cylindrical coordinates:

$$\frac{1}{r}\frac{\partial(r u_r)}{\partial r} + \frac{1}{r}\frac{\partial(u_\theta)}{\partial \theta} + \frac{\partial(u_z)}{\partial z} = 0$$

Topic No. 125

THE CONTINUITY EQUATION: Examples of Application of the Continuity Equation

Example: Design of a compressible converging duct

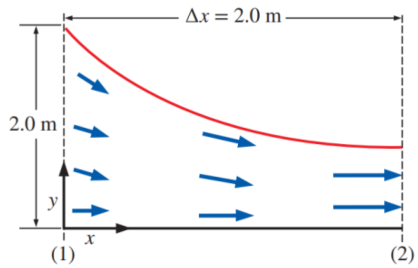
A two-dimensional converging duct is being designed for a high-speed wind tunnel. The bottom wall of the duct is to be flat and horizontal, and the top wall is to be curved in such a way that the axial wind speed u increases approximately linearly from $u_1 = 100$ m/s at section (1) to $u_2 = 300$ m/s at section

(2) (Fig. 9–12). Meanwhile, the air density ρ is to decrease approximately linearly from $\rho_1 = 1.2$ kg/m^3 at section (1) to $\rho_2 = 0.85$ kg/m^3 at section (2). The converging duct is 2.0 m long and is 2.0 m high at section (1). (*a*) Predict the *y-component* of velocity, $v(x, y)$, in the duct. (*b*) Plot the approximate shape of the duct, ignoring friction on the walls. (*c*) How high should the duct be at section (2), the exit of the duct?

SOLUTION For given velocity component u and density ρ, we are to predict velocity component v, plot an approximate shape of the duct, and predict its height at the duct exit.

Assumptions **1** The flow is steady and two-dimensional in the xy-plane. **2** Friction on the walls is ignored. **3** Axial velocity u increases linearly with x, and density ρ decreases linearly with x.

Properties The fluid is air at room temperature (25°C). The speed of sound is about 346 m/s, so the flow is subsonic, but compressible.



Converging duct, designed for a high speed wind tunnel (not to scale).

Topic No. 126

Examples of Application of the Continuity Equation

Example: Design of a compressible converging duct

Analysis (*a*) We write expressions for u and ρ, forcing them to be linear in x,

$$u = u_1 + C_u x \quad \text{where} \quad C_u = \frac{u_2 - u_1}{\Delta x} = \frac{(300 - 100)\text{ m/s}}{2.0\text{ m}} = 100\text{ s}^{-1} \qquad \textbf{(1)}$$

and

$$\rho = \rho_1 + C_\rho x \quad \text{where} \quad C_\rho = \frac{\rho_2 - \rho_1}{\Delta x} = \frac{(0.85 - 1.2)\text{ kg/m}^3}{2.0\text{ m}} \qquad \textbf{(2)}$$

$$= -0.175\text{ kg/m}^4$$

The steady continuity equation (Eq. 9–14) for this two-dimensional compressible flow simplifies to

$$\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \underbrace{\cancel{\frac{\partial(\rho w)}{\partial z}}}_{0\text{ (2-D)}} = 0 \quad \rightarrow \quad \frac{\partial(\rho v)}{\partial y} = -\frac{\partial(\rho u)}{\partial x} \tag{3}$$

Substituting Eqs. 1 and 2 into Eq. 3 and noting that C_u and C_ρ are constants,

$$\frac{\partial(\rho v)}{\partial y} = -\frac{\partial[(\rho_1 + C_\rho x)(u_1 + C_u x)]}{\partial x} = -(\rho_1 C_u + u_1 C_\rho) - 2C_u C_\rho x$$

Integration with respect to y gives

$$\rho v = -(\rho_1 C_u + u_1 C_\rho)y - 2C_u C_\rho xy + f(x) \tag{4}$$

Note that since the integration is a *partial* integration, we have added an arbitrary function of x instead of simply a constant of integration. Next, we apply boundary conditions. We argue that since the bottom wall is flat and horizontal, v must equal zero at $y = 0$ for any x. This is possible only if $f(x) = 0$. Solving Eq. 4 for v gives

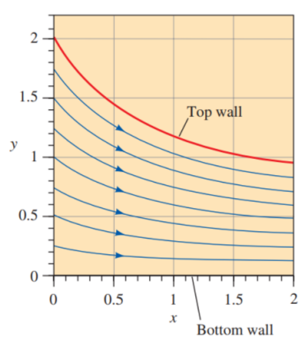
$$v = \frac{-(\rho_1 C_u + u_1 C_\rho)y - 2C_u C_\rho xy}{\rho} \quad \rightarrow \quad \boldsymbol{v = \frac{-(\rho_1 C_u + u_1 C_\rho)y - 2C_u C_\rho xy}{\rho_1 + C_\rho x}} \tag{5}$$

(*b*) Using Eqs. 1 and 5 and the technique described in Chap. 4, we plot several streamlines between $x = 0$ and $x = 2.0$ m in Fig. 9–13. The streamline starting at $x = 0$, $y = 2.0$ m approximates the top wall of the duct.

(*c*) At section (2), the top streamline crosses $y = 0.941$ m at $x = 2.0$ m. Thus, the predicted height of the duct at section (2) is **0.941 m**.

Discussion You can verify that the combination of Eqs. 1, 2, and 5 satisfies the continuity equation. However, this alone does not guarantee that the density and velocity components will actually *follow* these equations if the duct were to be built as designed here. The actual flow depends on the *pressure drop* between sections (1) and (2); only one unique pressure drop can yield the desired flow acceleration. Temperature may also change considerably in this kind of compressible flow in which the air accelerates toward sonic speeds.

Streamlines for the converging duct



Topic No. 127

Examples of Application of the Continuity Equation

Example: Incompressibility of an Unsteady Two-Dimensional Flow

Consider the velocity field of Example 4–5—an unsteady, two-dimensional velocity field given by $\vec{V} = (u, v) = (0.5 + 0.8x)\vec{i} + [1.5 + 2.5 \sin(\omega t) - 0.8y]\vec{j}$, where angular frequency ω is equal to 2π rad/s (a physical frequency of 1 Hz). Verify that this flow field can be approximated as incompressible.

SOLUTION We are to verify that a given velocity field is incompressible.
Assumptions **1** The flow is two-dimensional, implying no z-component of velocity and no variation of u or v with z.
Analysis The components of velocity in the x- and y-directions, respectively, are

$$u = 0.5 + 0.8x \qquad \text{and} \qquad v = 1.5 + 2.5 \sin(\omega t) - 0.8y$$

If the flow is incompressible, Eq. 9–16 must apply. More specifically, in Cartesian coordinates Eq. 9–17 must apply. Let's check:

$$\underbrace{\frac{\partial u}{\partial x}}_{0.8} + \underbrace{\frac{\partial v}{\partial y}}_{-0.8} + \underbrace{\cancel{\frac{\partial w}{\partial z}}}_{0 \text{ since 2-D}} = 0 \quad \rightarrow \quad 0.8 - 0.8 = 0$$

So we see that the incompressible continuity equation is indeed satisfied at any instant in time, and **this flow field may be approximated as incompressible**.
Discussion Although there is an unsteady term in v, it has no y-derivative and drops out of the continuity equation.

Topic No. 128

Examples of Application of the Continuity Equation

Three-Dimensional Flow

Example: Finding a Missing Velocity Component

Two velocity components of a steady, incompressible, three-dimensional flow field are known, namely, $u = ax^2 + by^2 + cz^2$ and $w = axz + byz^2$, where a, b, and c are constants. The y velocity component is missing (Fig. 9–14). Generate an expression for v as a function of x, y, and z.

SOLUTION We are to find the y-component of velocity, v, using given expressions for u and w.
Assumptions **1** The flow is steady. **2** The flow is incompressible.
Analysis Since the flow is steady and incompressible, and since we are working in Cartesian coordinates, we apply Eq. 9–17 to the flow field,

Condition for incompressibility:

$$\frac{\partial v}{\partial y} = -\underbrace{\frac{\partial u}{\partial x}}_{2ax} - \underbrace{\frac{\partial w}{\partial z}}_{ax+2byz} \rightarrow \frac{\partial v}{\partial y} = -3ax - 2byz$$

Next we integrate with respect to y. Since the integration is a *partial* integration, we add some arbitrary function of x and z instead of a simple constant of integration.

Solution: $$v = -3axy - by^2z + f(x,z)$$

Discussion Any function $f(x,z)$ yields a v that satisfies the incompressible continuity equation, since there are no derivatives of v with respect to x or z in the continuity equation.

Topic No. 129

Examples of Application of the Continuity Equation

Example: Two-Dimensional, Incompressible, Vertical Flow

Consider a two-dimensional, incompressible flow in cylindrical coordinates; the tangential velocity component is $u_\theta = K/r$, where K is a constant. This represents a class of vortical flows. Generate an expression for the other velocity component, u_r.

SOLUTION For a given tangential velocity component, we are to generate an expression for the radial velocity component.

Assumptions **1** The flow is two-dimensional in the xy- ($r\theta$-) plane (velocity is not a function of z, and $u_z = 0$ everywhere). **2** The flow is incompressible.

Analysis The incompressible continuity equation (Eq. 9–18) for this two-dimensional case simplifies to

$$\frac{1}{r}\frac{\partial (ru_r)}{\partial r} + \frac{1}{r}\frac{\partial u_\theta}{\partial \theta} + \underbrace{\cancel{\frac{\partial u_z}{\partial z}}}_{0\ (2\text{-D})} = 0 \quad \rightarrow \quad \frac{\partial (ru_r)}{\partial r} = -\frac{\partial u_\theta}{\partial \theta} \tag{1}$$

The given expression for u_θ is not a function of θ, and therefore Eq. 1 reduces to

$$\frac{\partial (ru_r)}{\partial r} = 0 \quad \rightarrow \quad ru_r = f(\theta, t) \tag{2}$$

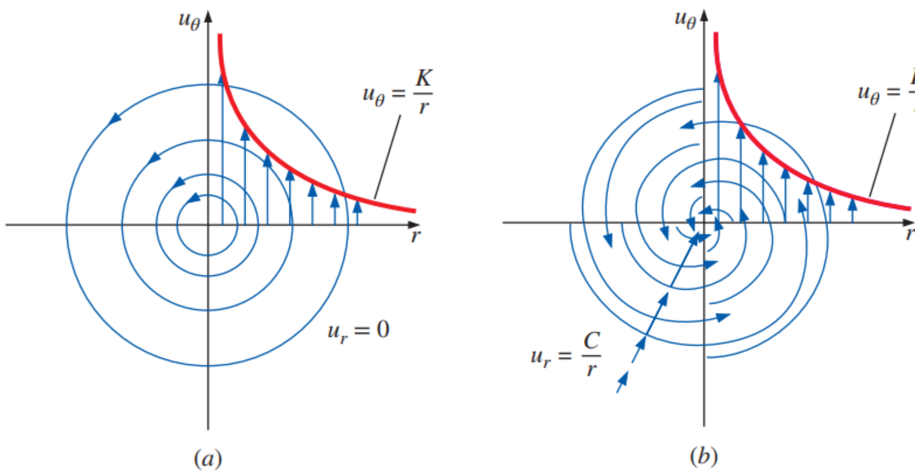
where we have introduced an arbitrary function of θ and t instead of a constant of integration, since we performed a *partial* integration with respect to r. Solving for u_r,

$$u_r = \frac{f(\theta, t)}{r} \tag{3}$$

Thus, *any radial velocity component of the form given by Eq. 3 yields a two-dimensional, incompressible velocity field that satisfies the continuity equation.*

We discuss some specific cases. The simplest case is when $f(\theta, t) = 0$ ($u_r = 0$, $u_\theta = K/r$). This yields the **line vortex** discussed in Chap. 4, as sketched in Fig. 9–15*a*. Another simple case is when $f(\theta, t) = C$, where C is a constant. This yields a radial velocity whose magnitude decays as $1/r$. For negative C, imagine a spiraling line vortex/sink flow, in which fluid elements not only revolve around the origin, but get sucked into a sink at the origin (actually a line sink along the z-axis). This is illustrated in Fig. 9–15*b*.

Discussion Other more complicated flows can be obtained by setting $f(\theta, t)$ to some other function. For any function $f(\theta, t)$, the flow satisfies the two-dimensional, incompressible continuity equation at a given instant in time.



Streamlines and velocity profiles for (*a*) a line vortex flow and (*b*) a spiraling line vortex/sink flow.

Topic No. 130

Examples of Application of the Continuity Equation

Example: Conditions for Incompressible Flow

Consider a steady velocity field given by $\vec{V} = (u, v, w) = a(x^2y + y^2)\vec{i} + bxy^2\vec{j} + cx\vec{k}$, where a, b, and c are constants. Under what conditions is this flow field incompressible?

SOLUTION We are to determine a relationship between constants a, b, and c that ensures incompressibility.

Assumptions **1** The flow is steady. **2** The flow is incompressible (under certain constraints to be determined).

Analysis We apply Eq. 9–17 to the given velocity field,

$$\underbrace{\frac{\partial u}{\partial x}}_{2axy} + \underbrace{\frac{\partial v}{\partial y}}_{2bxy} + \underbrace{\cancel{\frac{\partial w}{\partial z}}}_{0} = 0 \quad \rightarrow \quad 2axy + 2bxy = 0$$

Thus to guarantee incompressibility, constants a and b must be equal in magnitude but opposite in sign.

Condition for incompressibility: $\qquad \boldsymbol{a = -b}$

Discussion If a were not equal to $-b$, this might still be a valid flow field, but density would have to vary with location in the flow field. In other words, the flow would be *compressible,* and Eq. 9–14 would need to be satisfied in place of Eq. 9–17.

Topic No. 131

The Stream Function

The Stream Function in Cartesian Coordinates

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

A clever variable transformation enables us to rewrite above equation in terms of *one* dependent variable (ψ) instead of *two* dependent variables (u and v). We define the **stream function** ψ

Incompressible, two-dimensional stream function in Cartesian coordinates:

$$u = \frac{\partial \psi}{\partial y} \quad \text{and} \quad v = -\frac{\partial \psi}{\partial x}$$

The stream function and the corresponding velocity potential function were first introduced by the Italian mathematician Joseph Louis Lagrange (1736–1813). Substitution of above equations yields

$$\frac{\partial}{\partial x}\left(\frac{\partial \psi}{\partial y}\right) + \frac{\partial}{\partial y}\left(-\frac{\partial \psi}{\partial x}\right) = \frac{\partial^2 \psi}{\partial x\, \partial y} - \frac{\partial^2 \psi}{\partial y\, \partial x} = 0$$

which is identically satisfied for any smooth function $\psi(x, y)$, because the order of differentiation (y then x versus x then y) is irrelevant.

$$\textit{Along a streamline:} \quad \frac{dy}{dx} = \frac{v}{u} \quad \rightarrow \quad \underbrace{-v}_{\partial\psi/\partial x}\, dx + \underbrace{u}_{\partial\psi/\partial y}\, dy = 0$$

Along a streamline:
$$\frac{\partial \psi}{\partial x}\, dx + \frac{\partial \psi}{\partial y}\, dy = 0$$

Total change of ψ:
$$d\psi = \frac{\partial \psi}{\partial x}\, dx + \frac{\partial \psi}{\partial y}\, dy$$

By comparing above equations we see that $d\psi = 0$ along a streamline; thus we have proven the statement that ψ is constant along streamlines.

Curves of constant ψ are streamlines of the flow.

Stream Function

- 2-D, incompressible, Cartesian coordinates:

$$u = \frac{\partial \psi}{\partial y} \quad \text{and} \quad \upsilon = -\frac{\partial \psi}{\partial x}$$

- 2-D, incompressible, cylindrical coordinates:

$$u_r = \frac{1}{r}\frac{\partial \psi}{\partial \theta} \quad \text{and} \quad u_\theta = -\frac{\partial \psi}{\partial r}$$

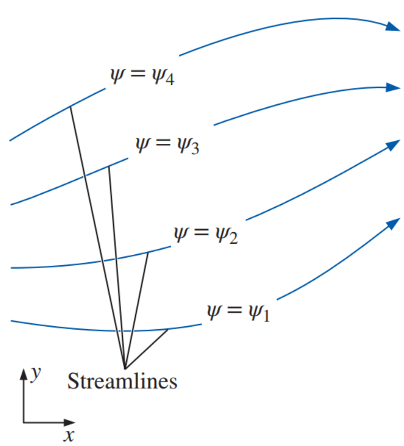
- Axisymmetric, incompressible, cylindrical coordinates:

$$u_r = -\frac{1}{r}\frac{\partial \psi}{\partial z} \quad \text{and} \quad u_z = \frac{1}{r}\frac{\partial \psi}{\partial r}$$

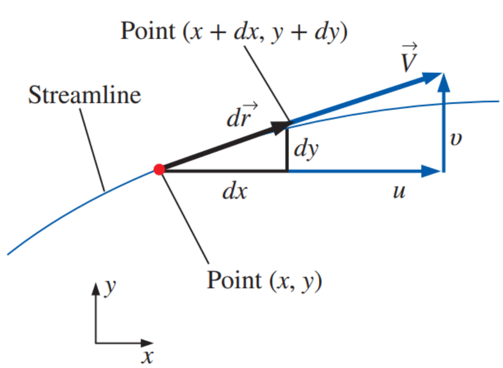
- 2-D, compressible, Cartesian coordinates:

$$\rho u = \frac{\partial \psi_\rho}{\partial y} \quad \text{and} \quad \rho \upsilon = -\frac{\partial \psi_\rho}{\partial x}$$

There are several definitions of the stream function, depending on the type of flow under consideration as well as the coordinate system being used.



Curves of constant stream function represent streamlines of the flow.



Arc length $d\vec{r} = (dx, dy)$ and local velocity vector $\vec{V} = (u, \upsilon)$ along a two-dimensional streamline in the *xy*-plane.

Topic No. 132

The Stream Function

Example: Calculation of the Velocity Field from the Stream Function

A steady, two-dimensional, incompressible flow field in the xy-plane has a stream function given by $\psi = ax^3 + by + cx$, where a, b, and c are constants: $a = 0.50\ (\text{m·s})^{-1}$, $b = -2.0$ m/s, and $c = -1.5$ m/s. (a) Obtain expressions for velocity components u and υ. (b) Verify that the flow field satisfies the incompressible continuity equation. (c) Plot several streamlines of the flow in the upper-right quadrant.

SOLUTION For a given stream function, we are to calculate the velocity components, verify incompressibility, and plot flow streamlines.

Assumptions **1** The flow is steady. **2** The flow is incompressible (this assumption is to be verified). 3 The flow is two-dimensional in the xy-plane, implying that $w = 0$ and neither u nor υ depend on z.

Analysis (a) We use Eq. 9–20 to obtain expressions for u and υ by differentiating the stream function,

$$u = \frac{\partial \psi}{\partial y} = \boldsymbol{b} \quad \text{and} \quad \upsilon = -\frac{\partial \psi}{\partial \text{x}} = \boldsymbol{-3ax^2 - c}$$

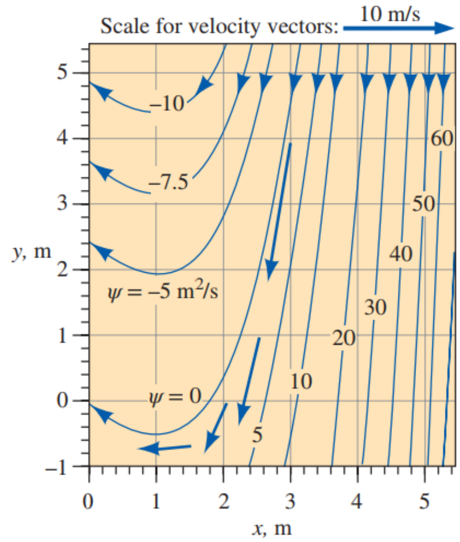
(b) Since u is not a function of x, and υ is not a function of y, we see immediately that the two-dimensional, incompressible continuity equation (Eq. 9–19) is satisfied. In fact, since ψ is smooth in x and y, the two-dimensional, incompressible continuity equation in the xy-plane is automatically satisfied by the very definition of ψ. We conclude that **the flow is indeed incompressible**.

(c) To plot streamlines, we solve the given equation for either y as a function of x and ψ, or x as a function of y and ψ. In this case, the former is easier, and we have

Equation for a streamline: $$y = \frac{\psi - ax^3 - cx}{b}$$

This equation is plotted in Fig. 9–20 for several values of ψ, and for the provided values of a, b, and c. The flow is nearly straight down at large values of x, but veers upward for $x < 1$ m.

Discussion You can verify that $\upsilon = 0$ at $x = 1$ m. In fact, υ is negative for $x > 1$ m and positive for $x < 1$ m. The direction of the flow can also be determined by picking an arbitrary point in the flow, say ($x = 3$ m, $y = 4$ m), and calculating the velocity there. We get $u = -2.0$ m/s and $\upsilon = -12.0$ m/s at this point, either of which shows that fluid flows to the lower left in this region of the flow field. For clarity, the velocity vector at this point is also plotted in Fig. 9–20; it is clearly parallel to the streamline near that point. Velocity vectors at three other locations are also plotted.



Streamlines for the velocity field of example; the value of constant ψ is indicated for each streamline, and velocity vectors are shown at four locations.

Topic No. 133

The Stream Function

Example: Calculation of Stream Function for a Known Velocity Field

Consider a steady, two-dimensional, incompressible velocity field with $u = ax + b$ and $v = -ay + cx$, where a, b, and c are constants: $a = 0.50\ s^{-1}$, $b = 1.5$ m/s, and $c = 0.35\ \text{s}^{-1}$. Generate an expression for the stream function and plot some streamlines of the flow in the upper-right quadrant.

SOLUTION For a given velocity field we are to generate an expression for ψ and plot several streamlines for given values of constants a, b, and c.

Assumptions **1** The flow is steady. **2** The flow is incompressible. **3** The flow is two-dimensional in the xy-plane, implying that $w = 0$ and neither u nor v depend on z.

Analysis We start by picking one of the two parts of Eq. 9–20 that define the stream function (it doesn't matter which part we choose—the solution will be identical).

$$\frac{\partial \psi}{\partial y} = u = ax + b$$

Next we integrate with respect to y, noting that this is a *partial* integration, so we add an arbitrary function of the other variable, x, rather than a constant of integration,

$$\psi = axy + by + g(x) \tag{1}$$

Now we choose the other part of Eq. 9–20, differentiate Eq. 1, and rearrange as follows:

$$v = -\frac{\partial \psi}{\partial x} = -ay - g'(x) \qquad \textbf{(2)}$$

where $g'(x)$ denotes dg/dx since g is a function of only one variable, x. We now have two expressions for velocity component v, the equation given in the problem statement and Eq. 2. We equate these and integrate with respect to x to find $g(x)$,

$$v = -ay + cx = -ay - g'(x) \quad \rightarrow \quad g'(x) = -cx \quad \rightarrow \quad g(x) = -c\frac{x^2}{2} + C \qquad \textbf{(3)}$$

Note that here we have added an arbitrary constant of integration C since g is a function of x only. Finally, substituting Eq. 3 into Eq. 1 yields the final expression for ψ,

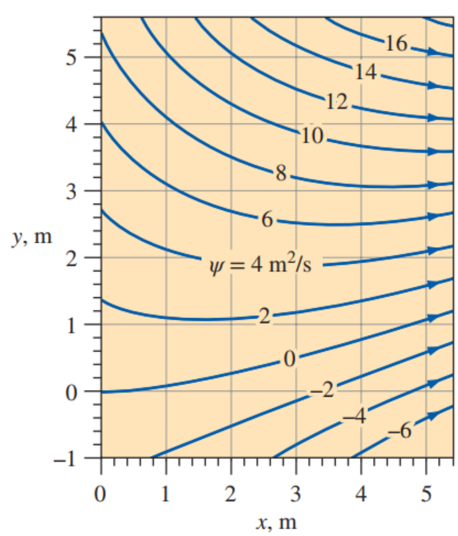
Solution:
$$\boldsymbol{\psi = axy + by - c\frac{x^2}{2} + C} \qquad \textbf{(4)}$$

To plot the streamlines, we note that Eq. 4 represents a *family* of curves, one unique curve for each value of the constant $(\psi - C)$. Since C is arbitrary, it is common to set it equal to zero, although it can be set to any desired value. For simplicity we set $C = 0$ and solve Eq. 4 for y as a function of x, yielding

Equation for streamlines:
$$y = \frac{\psi + cx^2/2}{ax + b} \qquad \textbf{(5)}$$

For the given values of constants a, b, and c, we plot Eq. 5 for several values of ψ in Fig. 9–21; these curves of constant ψ are streamlines of the flow. From Fig. 9–21 we see that this is a smoothly converging flow in the upper-right quadrant.

Discussion It is always good to check your algebra. In this example, you should substitute Eq. 4 into Eq. 9–20 to verify that the correct velocity components are obtained.



Streamlines for the velocity field of example; the value of constant ψ is indicated for each streamline.

Topic No. 134

The Stream Function in Cylindrical Coordinates

$$\frac{\partial(ru_r)}{\partial r} + \frac{\partial(u_\theta)}{\partial \theta} = 0$$

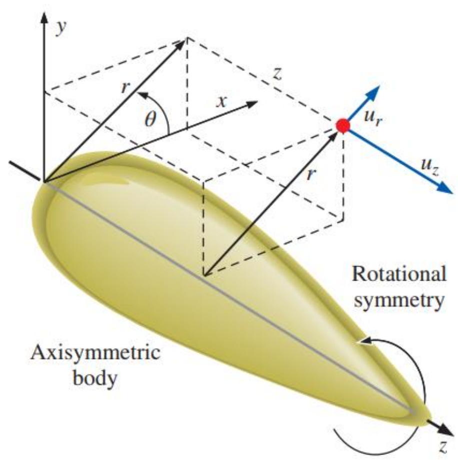
Incompressible, planar stream function in cylindrical coordinates:

$$u_r = \frac{1}{r}\frac{\partial \psi}{\partial \theta} \quad \text{and} \quad u_\theta = -\frac{\partial \psi}{\partial r}$$

$$\frac{1}{r}\frac{\partial(ru_r)}{\partial r} + \frac{\partial(u_z)}{\partial z} = 0$$

Incompressible, axisymmetric stream function in cylindrical coordinates:

$$u_r = -\frac{1}{r}\frac{\partial \psi}{\partial z} \quad \text{and} \quad u_z = \frac{1}{r}\frac{\partial \psi}{\partial r}$$



Flow over an axisymmetric body in cylindrical coordinates with rotational symmetry about the *z*-axis; neither the geometry nor the velocity field depend on θ, and $u_\theta = 0$.

Example: Stream Function in Cylindrical Coordinates

Consider a line vortex, defined as steady, planar, incompressible flow in which the velocity components are $u_r = 0$ and $u_\theta = K/r$, where K is a constant. This flow is represented in Fig. 9–15*a*. Derive an expression for the stream function $\psi(r, \theta)$, and prove that the streamlines are circles.

SOLUTION For a given velocity field in cylindrical coordinates, we are to derive an expression for the stream function and show that the streamlines are circular.

Assumptions 1 The flow is steady. 2 The flow is incompressible. 3 The flow is planar in the $r\theta$-plane.

Analysis We use the definition of stream function given by Eq. 9–27. We can choose either component to start with; we choose the tangential component,

$$\frac{\partial \psi}{\partial r} = -u_\theta = -\frac{K}{r} \quad \rightarrow \quad \psi = -K \ln r + f(\theta) \tag{1}$$

Now we use the other component of Eq. 9–27,

$$u_r = \frac{1}{r}\frac{\partial \psi}{\partial \theta} = \frac{1}{r} f'(\theta) \tag{2}$$

where the prime denotes a derivative with respect to θ. By equating u_r from the given information to Eq. 2, we see that

$$f'(\theta) = 0 \quad \rightarrow \quad f(\theta) = C$$

where C is an arbitrary constant of integration. Equation 1 is thus

Solution: $$\boldsymbol{\psi = -K \ln r + C} \tag{3}$$

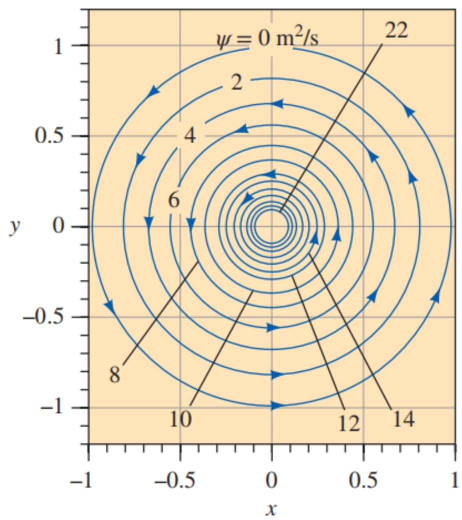
Finally, we see from Eq. 3 that curves of constant ψ are produced by setting r to a constant value. Since curves of constant r are circles by definition, **streamlines (curves of constant ψ) must therefore be circles about the origin, as in Fig. 9–15*a***.

For given values of C and ψ, we solve Eq. 3 for r to plot the streamlines,

Equation for streamlines: $$r = e^{-(\psi - C)/K} \tag{4}$$

For $K = 10\ m^2/s$ and $C = 0$, streamlines from $\psi = 0$ to 22 are plotted in Fig. 9–28.

Discussion Notice that for a uniform increment in the value of ψ, the streamlines get closer and closer together near the origin as the tangential velocity increases. This is a direct result of the statement that the difference in the value of ψ from one streamline to another is equal to the volume flow rate per unit width between the two streamlines.



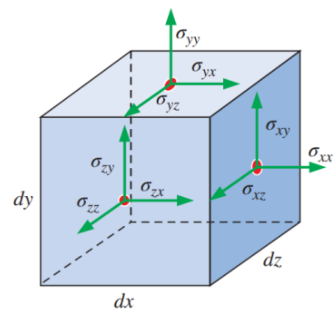
Streamlines for the velocity field of example, with $K = 10\ m^2/s$ and $C = 0$; the value of constant ψ is indicated for several streamlines.

Topic No. 135

The Differential Linear Momentum Equation Cauchy's Equation

- Through application of the Reynolds transport theorem, we have the general expression for the linear momentum equation as applied to a control volume,

$$\sum \vec{F} = \int_{CV} \rho\vec{g}\, dV + \int_{CS} \sigma_{ij}\cdot\vec{n}\, dA = \int_{CV} \frac{\partial}{\partial t}(\rho\vec{V})\, dV + \int_{CS} (\rho\vec{V})\vec{V}\cdot\vec{n}\, dA$$ ___(135-a)



Positive components of the stress tensor σ_{ij} in Cartesian coordinates on the positive (right, top, and front) faces of an infinitesimal rectangular control volume. The red dots indicate the center of each face. Positive components on the negative (left, bottom, and back) faces are in the opposite direction of those shown here.

- For the special case of flow with well-defined inlets and outlets, above equation is simplified as follows:

$$\sum \vec{F} = \sum \vec{F}_{body} + \sum \vec{F}_{surface} = \int_{CV} \frac{\partial}{\partial t}(\rho\vec{V})\, dV + \sum_{out} \beta\dot{m}\vec{V} - \sum_{in} \beta\dot{m}\vec{V}$$ ___(135-b)

Where $\vec{V}$ in the last two terms is taken as the average velocity at an inlet or outlet, and β is the momentum flux correction factor. Thus the total force acting on the control volume is equal to the rate at which momentum changes within the control volume plus the rate at which momentum flows out of the control volume minus the rate at which momentum regardless of its size.

Derivation Using the Divergence Theorem

- The most straightforward (and most elegant) way to derive the differential form of the momentum equation is to apply the divergence theorem:

$$\int_V \vec{\nabla}\cdot G_{ij}\, dV = \oint_A G_{ij}\cdot\vec{n}\, dA$$

$$\text{or} \int_V \vec{\nabla}\cdot \vec{\vec{G}}\, dV = \oint_A \vec{\vec{G}}\cdot\vec{n}\, dA$$

An extended form of the divergence theorem is useful not only for vectors, but also for tensors. In the equation, G_{ij} (or $\vec{\vec{G}}$) is a second-order tensor, V is a volume, and A is the surface area that encloses and defines the volume.

Topic No. 136

Derivation Using the Divergence Theorem

Specifically, if we replace G_{ij} in the extended divergence theorem with the quantity $(\rho\vec{V})\vec{V}$, a second-order tensor, the last term in Eq. (135-a) becomes

$$\int_{CS} (\rho\vec{V})\vec{V}\cdot\vec{n}\,dA = \int_{CV} \vec{\nabla}\cdot(\rho\vec{V}\vec{V})\,dV$$ ___(135-c)

Where $\vec{V}\vec{V}$ is a vector product called the outer product of the velocity vector with itself. (The outer product of two vectors is not the same as the inner or dot product, nor is it the same as the cross product of the two vectors.) Similarly, if we replace G_{ij} (in Divergence Theorem) by the stress tensor σ_{ij}, the second term on the left-hand side of Eq. (135-a) becomes

$$\int_{CS} \sigma_{ij}\cdot\vec{n}\,dA = \int_{CV} \vec{\nabla}\cdot\sigma_{ij}\,dV$$ ___(135-d)

Thus, the two surface integrals of Eq. (135-a) become volume integrals by applying Eqs. (135-c) and (135-d). We combine and rearrange the terms, and rewrite Eq. (135-a) as

$$\int_{CV} \left[\frac{\partial}{\partial t}(\rho\vec{V}) + \vec{\nabla}\cdot(\rho\vec{V}\vec{V}) - \rho\vec{g} - \vec{\nabla}\cdot\sigma_{ij}\right] dV = 0$$ ___(135-e)

The above equation can only be zero if the integrand is zero. Hence, we have a general differential equation for linear momentum, known as **Cauchy's equation**,

Cauchy's equation: $$\frac{\partial}{\partial t}(\rho\vec{V}) + \vec{\nabla}\cdot(\rho\vec{V}\vec{V}) = \rho\vec{g} + \vec{\nabla}\cdot\sigma_{ij}$$ ___(135-f)

Cauchy's equation is a differential form of the linear momentum equation. It applies to any type of fluid.

Equation (135-f) is named in honor of the French engineer and mathematician Augustin Louis de Cauchy (1789–1857). It is valid for compressible as well as incompressible flow since we have not made any assumptions about incompressibility. It is valid at any point in the flow domain. Note that Eq. (135-f) is a *vector* equation, and thus represents three scalar equations, one for each coordinate axis in three-dimensional problems.

Alternative form of Cauchy's equation:

$$\rho\left[\frac{\partial \vec{V}}{\partial t} + (\vec{V}\cdot\vec{\nabla})\vec{V}\right] = \rho\frac{D\vec{V}}{Dt} = \rho\vec{g} + \vec{\nabla}\cdot\sigma_{ij} \qquad \text{\_\_\_(135-g)}$$

Topic No. 137

The Navier-Stokes Equation: Introduction

- Cauchy's equation (or its alternative form) is not very useful to us as is, because the stress tensor σ_{ij} contains nine components, six of which are independent (because of symmetry). Thus, in addition to density and the three velocity components, there are six additional unknowns, for a total of 10 unknowns. (In Cartesian coordinates the unknowns are ρ, u, v, w, σ_{xx}, σ_{xy}, σ_{xz}, σ_{yy}, σ_{yz}, and σ_{zz}.)
- We have discussed only four equations so far—continuity (one equation) and Cauchy's equation (three equations). We need six more equations which we obtain from components of the stress tensor in terms of the velocity field and pressure field.
- For this we separate the pressure stresses and the viscous stresses. When a fluid is at rest, the only stress acting at any surface of any fluid element is pressure P, which always acts inward and normal to the surface. Thus, regardless of the orientation of the coordinate axes, for a fluid at rest the stress tensor reduces to

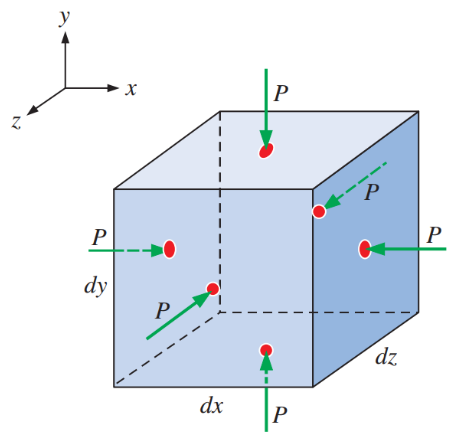
Fluid at rest:

$$\sigma_{ij} = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{pmatrix} = \begin{pmatrix} -P & 0 & 0 \\ 0 & -P & 0 \\ 0 & 0 & -P \end{pmatrix}$$

Moving fluids:

$$\sigma_{ij} = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{pmatrix} = \begin{pmatrix} -P & 0 & 0 \\ 0 & -P & 0 \\ 0 & 0 & -P \end{pmatrix} + \begin{pmatrix} \tau_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \tau_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \tau_{zz} \end{pmatrix}$$

τ_{ij}, called the **viscous stress tensor** or the **deviatoric stress tensor**. Mathematically, we have not helped the situation because we have replaced the six unknown components of σ_{ij} with six unknown components of τ_{ij}, and have added *another* unknown, pressure P. Fortunately; however, there are constitutive equations that express τ_{ij} in terms of the velocity field and measurable fluid properties such as viscosity.



For fluids at rest, the only stress on a fluid element is the hydrostatic pressure, which always acts inward and normal to any surface. Note that we are ignoring gravity in this case; otherwise pressure would increase in the direction of the gravitational acceleration.

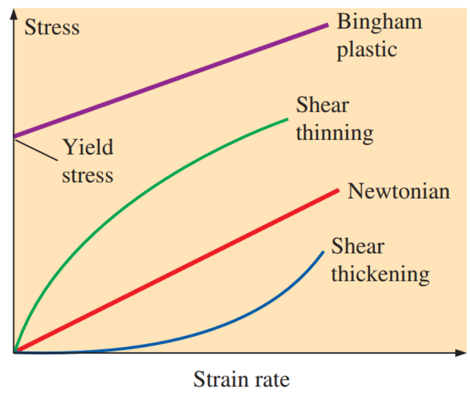
Mechanical pressure is the mean normal stress acting inwardly on a fluid element. It is therefore also called **mean pressure.**

Mechanical pressure: $$P_m = -\frac{1}{3}(\sigma_{xx} + \sigma_{yy} + \sigma_{zz})$$

Topic No. 138

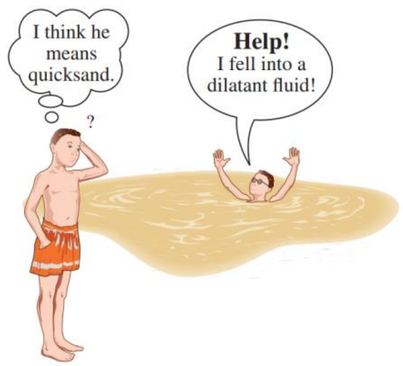
Newtonian versus Non-Newtonian Fluids

- The study of the deformation of flowing fluids is called **rheology.**
- **Newtonian fluids**, defined as fluids for which the stress tensor is linearly proportional to the strain rate tensor.
- Fluids for which the stress tensor is not linearly related to the strain rate tensor are called **non-Newtonian fluids.**
- A fluid that returns (partially) to its original shape after the applied stress is released is called **viscoelastic.**



Rheological behavior of fluids—stress as a function of strain rate.

- Some non-Newtonian fluids are called **shear thinning fluids** or **pseudo plastic fluids**, because the more the fluid is sheared, the less viscous it becomes.
- **Plastic fluids** are those in which the shear thinning effect is extreme.
- In some fluids a finite stress called the **yield stress** is required before the fluid begins to flow at all; such fluids are called **Bingham plastic fluids**.
- **Shear thickening fluids** or **dilatant fluids**; the more the fluid is sheared, the more viscous it becomes.

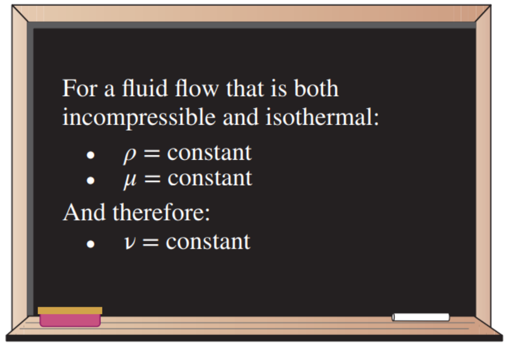


When someone falls into quicksand (a dilatant fluid), the faster he tries to move, the more viscous the fluid becomes.

Topic No. 139

Derivation of the Navier–Stokes Equation for Incompressible, Isothermal Flow

The incompressible flow approximation implies constant density, and the isothermal approximation implies constant viscosity.



- Viscous stress tensor for an incompressible Newtonian fluid with constant properties:

$$\tau_{ij} = 2\mu\varepsilon_{ij}$$

Where ε_{ij} is the strain rate tensor defined in above equation shows that stress is linearly proportional to strain. In Cartesian coordinates, the nine components of the viscous stress tensor are listed, only six of which are independent due to symmetry:

$$\tau_{ij} = \begin{pmatrix} \tau_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \tau_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \tau_{zz} \end{pmatrix} = \begin{pmatrix} 2\mu \dfrac{\partial u}{\partial x} & \mu\left(\dfrac{\partial u}{\partial y} + \dfrac{\partial v}{\partial x}\right) & \mu\left(\dfrac{\partial u}{\partial z} + \dfrac{\partial w}{\partial x}\right) \\ \mu\left(\dfrac{\partial v}{\partial x} + \dfrac{\partial u}{\partial y}\right) & 2\mu \dfrac{\partial v}{\partial y} & \mu\left(\dfrac{\partial v}{\partial z} + \dfrac{\partial w}{\partial y}\right) \\ \mu\left(\dfrac{\partial w}{\partial x} + \dfrac{\partial u}{\partial z}\right) & \mu\left(\dfrac{\partial w}{\partial y} + \dfrac{\partial v}{\partial z}\right) & 2\mu \dfrac{\partial w}{\partial z} \end{pmatrix}$$

$$\sigma_{ij} = \begin{pmatrix} -P & 0 & 0 \\ 0 & -P & 0 \\ 0 & 0 & -P \end{pmatrix} + \begin{pmatrix} 2\mu \dfrac{\partial u}{\partial x} & \mu\left(\dfrac{\partial u}{\partial y} + \dfrac{\partial v}{\partial x}\right) & \mu\left(\dfrac{\partial u}{\partial z} + \dfrac{\partial w}{\partial x}\right) \\ \mu\left(\dfrac{\partial v}{\partial x} + \dfrac{\partial u}{\partial y}\right) & 2\mu \dfrac{\partial v}{\partial y} & \mu\left(\dfrac{\partial v}{\partial z} + \dfrac{\partial w}{\partial y}\right) \\ \mu\left(\dfrac{\partial w}{\partial x} + \dfrac{\partial u}{\partial z}\right) & \mu\left(\dfrac{\partial w}{\partial y} + \dfrac{\partial v}{\partial z}\right) & 2\mu \dfrac{\partial w}{\partial z} \end{pmatrix}$$

Now we substitute above equation into the three Cartesian components of Cauchy's equation. Let's consider the x-component first. Equation becomes

$$\rho \frac{Du}{Dt} = -\frac{\partial P}{\partial x} + \rho g_x + 2\mu \frac{\partial^2 u}{\partial x^2} + \mu \frac{\partial}{\partial y}\left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\right) + \mu \frac{\partial}{\partial z}\left(\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z}\right)$$

Note that since pressure consists of a normal stress only, it contributes only one term to above equation. However, since the viscous stress tensor consists of both normal and shear stresses, it contributes three terms.

(This is a direct result of taking the divergence of a second-order tensor, by the way.)

We note that as long as the velocity components are smooth functions of *x*, *y*, and *z*, the order of differentiation is irrelevant. For example, the first part of the last term in above equation can be rewritten as

$$\mu \frac{\partial}{\partial z}\left(\frac{\partial w}{\partial x}\right) = \mu \frac{\partial}{\partial x}\left(\frac{\partial w}{\partial z}\right)$$

After some clever rearrangement of the viscous terms

$$\begin{aligned}\rho \frac{Du}{Dt} &= -\frac{\partial P}{\partial x} + \rho g_x + \mu\left[\frac{\partial^2 u}{\partial x^2} + \frac{\partial}{\partial x}\frac{\partial u}{\partial x} + \frac{\partial}{\partial x}\frac{\partial v}{\partial y} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial}{\partial x}\frac{\partial w}{\partial z} + \frac{\partial^2 u}{\partial z^2}\right] \\ &= -\frac{\partial P}{\partial x} + \rho g_x + \mu\left[\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}\right) + \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}\right]\end{aligned}$$

The term in parentheses is zero because of the continuity equation for incompressible flow. We also recognize the last three terms as the **Laplacian** of velocity component *u* in Cartesian coordinates. Thus, we write the *x*-component of the momentum equation as

$$\rho \frac{Du}{Dt} = -\frac{\partial P}{\partial x} + \rho g_x + \mu \nabla^2 u$$

Similarly, the *y*- and *z*-components of the momentum equation reduce to

$$\rho \frac{Dv}{Dt} = -\frac{\partial P}{\partial y} + \rho g_y + \mu \nabla^2 v$$

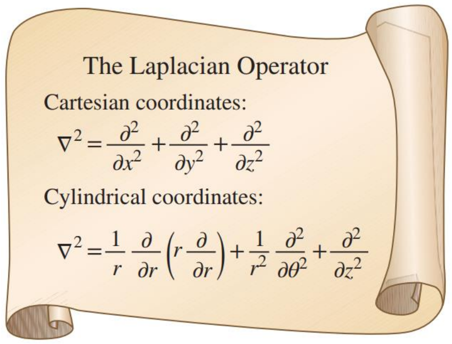
and

$$\rho \frac{Dw}{Dt} = -\frac{\partial P}{\partial z} + \rho g_z + \mu \nabla^2 w$$

respectively. Finally, we combine the three components into one vector equation; the result is the **Navier–Stokes equation** for incompressible flow with constant viscosity.

Incompressible Navier–Stokes equation:

$$\rho \frac{D\vec{V}}{Dt} = -\vec{\nabla} P + \rho \vec{g} + \mu \nabla^2 \vec{V}$$



The Laplacian operator, shown here in both Cartesian and cylindrical coordinates, appears in the viscous term of the incompressible Navier–Stokes equation.

Topic No. 140

Derivation of the Navier–Stokes Equation for Incompressible, Isothermal Flow

- The Navier–Stokes equation is the cornerstone of fluid mechanics. It may look harmless enough, but it is an unsteady, nonlinear, second-order, partial differential equation.
- Above equation has four unknowns (three velocity components and pressure), yet it represents only three equations (three components since it is a vector equation). The fourth equation is the incompressible continuity equation.
- We need to choose a coordinate system and expand the equations in that coordinate system.

The Navier–Stokes equation is the cornerstone of fluid mechanics.

Continuity and Navier–Stokes Equations in Cartesian Coordinates

The continuity equation and the Navier–Stokes equation are expanded in Cartesian coordinates (x, y, z) and (u, v, w):

Incompressible continuity equation:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

x-component of the incompressible Navier–Stokes equation:

$$\rho\left(\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z}\right) = -\frac{\partial P}{\partial x} + \rho g_x + \mu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}\right)$$

y-component of the incompressible Navier–Stokes equation:

$$\rho\left(\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z}\right) = -\frac{\partial P}{\partial y} + \rho g_y + \mu\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2}\right)$$

z-component of the incompressible Navier–Stokes equation:

$$\rho\left(\frac{\partial w}{\partial t} + u\frac{\partial w}{\partial x} + v\frac{\partial w}{\partial y} + w\frac{\partial w}{\partial z}\right) = -\frac{\partial P}{\partial z} + \rho g_z + \mu\left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2}\right)$$

Continuity and Navier–Stokes Equations in Cylindrical Coordinates

The continuity equation and the Navier–Stokes equation are expanded in cylindrical coordinates (r, θ, z) and (u_r, u_θ, u_z):

Incompressible continuity equation: $$\frac{1}{r}\frac{\partial(ru_r)}{\partial r} + \frac{1}{r}\frac{\partial(u_\theta)}{\partial \theta} + \frac{\partial(u_z)}{\partial z} = 0$$

r-component of the incompressible Navier–Stokes equation:

$$\rho\left(\frac{\partial u_r}{\partial t} + u_r\frac{\partial u_r}{\partial r} + \frac{u_\theta}{r}\frac{\partial u_r}{\partial \theta} - \frac{u_\theta^2}{r} + u_z\frac{\partial u_r}{\partial z}\right)$$

$$= -\frac{\partial P}{\partial r} + \rho g_r + \mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u_r}{\partial r}\right) - \frac{u_r}{r^2} + \frac{1}{r^2}\frac{\partial^2 u_r}{\partial \theta^2} - \frac{2}{r^2}\frac{\partial u_\theta}{\partial \theta} + \frac{\partial^2 u_r}{\partial z^2}\right]$$

θ-component of the incompressible Navier–Stokes equation:

$$\rho\left(\frac{\partial u_\theta}{\partial t} + u_r\frac{\partial u_\theta}{\partial r} + \frac{u_\theta}{r}\frac{\partial u_\theta}{\partial \theta} + \frac{u_r u_\theta}{r} + u_z\frac{\partial u_\theta}{\partial z}\right)$$

$$= -\frac{1}{r}\frac{\partial P}{\partial \theta} + \rho g_\theta + \mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u_\theta}{\partial r}\right) - \frac{u_\theta}{r^2} + \frac{1}{r^2}\frac{\partial^2 u_\theta}{\partial \theta^2} + \frac{2}{r^2}\frac{\partial u_r}{\partial \theta} + \frac{\partial^2 u_\theta}{\partial z^2}\right]$$

z-component of the incompressible Navier–Stokes equation:

$$\rho\left(\frac{\partial u_z}{\partial t} + u_r\frac{\partial u_z}{\partial r} + \frac{u_\theta}{r}\frac{\partial u_z}{\partial \theta} + u_z\frac{\partial u_z}{\partial z}\right)$$

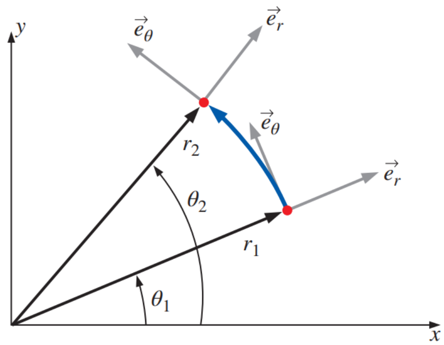
$$= -\frac{\partial P}{\partial z} + \rho g_z + \mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u_z}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u_z}{\partial \theta^2} + \frac{\partial^2 u_z}{\partial z^2}\right]$$

(This coupling effect is not present in Cartesian coordinates, and thus there are no “extra” terms in those equations.)
For completeness, the six independent components of the viscous stress tensor are listed here in cylindrical coordinates,

$$\tau_{ij} = \begin{pmatrix} \tau_{rr} & \tau_{r\theta} & \tau_{rz} \\ \tau_{\theta r} & \tau_{\theta\theta} & \tau_{\theta z} \\ \tau_{zr} & \tau_{z\theta} & \tau_{zz} \end{pmatrix}$$

$$= \begin{pmatrix} 2\mu\frac{\partial u_r}{\partial r} & \mu\left[r\frac{\partial}{\partial r}\left(\frac{u_\theta}{r}\right) + \frac{1}{r}\frac{\partial u_r}{\partial \theta}\right] & \mu\left(\frac{\partial u_r}{\partial z} + \frac{\partial u_z}{\partial r}\right) \\ \mu\left[r\frac{\partial}{\partial r}\left(\frac{u_\theta}{r}\right) + \frac{1}{r}\frac{\partial u_r}{\partial \theta}\right] & 2\mu\left(\frac{1}{r}\frac{\partial u_\theta}{\partial \theta} + \frac{u_r}{r}\right) & \mu\left(\frac{\partial u_\theta}{\partial z} + \frac{1}{r}\frac{\partial u_z}{\partial \theta}\right) \\ \mu\left(\frac{\partial u_r}{\partial z} + \frac{\partial u_z}{\partial r}\right) & \mu\left(\frac{\partial u_\theta}{\partial z} + \frac{1}{r}\frac{\partial u_z}{\partial \theta}\right) & 2\mu\frac{\partial u_z}{\partial z} \end{pmatrix}$$

The “extra” terms on both sides of the *r*- and θ-components of the Navier–Stokes equation arise because of the special nature of cylindrical coordinates. Namely, as we move in the θ-direction, the unit vector $\vec{e}_r$ also changes direction; thus the *r*- and θ-components are coupled. (This coupling effect is not present in Cartesian coordinates, and thus there are no “extra” terms in those equations.)



Unit vectors $\vec{e}_r$ and $\vec{e}_\theta$ in cylindrical coordinates are coupled: movement in the θ-direction causes $\vec{e}_r$ to change direction, and leads to extra terms in the *r*- and θ-components of the Navier–Stokes equation.

Topic No. 141

Differential Analysis of Fluid Flow Problems

There are two types of problems for which the differential equations (continuity and Navier–Stokes) are useful:

- Calculating the pressure field for a known velocity field
- Calculating both the velocity and pressure fields for a flow of known geometry and known boundary conditions

Three-Dimensional Incompressible Flow

Four variables or unknowns:

- Pressure P
- Three components of velocity $\vec{V}$

Four equations of motion:

- Continuity,

$$\vec{\nabla} \cdot \vec{V} = 0$$

- Three components of Navier–Stokes,

$$\rho \frac{D\vec{V}}{Dt} = -\vec{\nabla} P + \rho \vec{g} + \mu \nabla^2 \vec{V}$$

- A general three-dimensional but incompressible flow field with constant properties requires four equations to solve for four unknowns.

Calculation of the Pressure Field for a Known Velocity Field

- Examples involve calculation of the pressure field for a known velocity field.
- Since pressure does not appear in the continuity equation, we can theoretically generate a velocity field based solely on conservation of mass.
- However, since velocity appears in both the continuity equation and the Navier–Stokes equation, these two equations are coupled.
- Pressure appears in all three components of the Navier–Stokes equation, and thus the velocity and pressure fields are also coupled.
- This close coupling between velocity and pressure enables us to calculate the pressure field for a known velocity field.

Topic No. 142

Boundary Layer

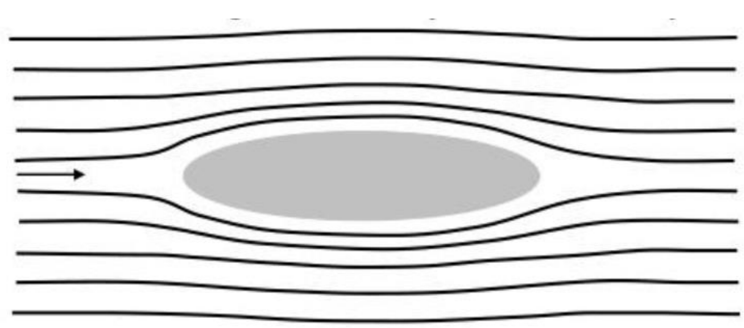
- The Navier-Stokes Equations
- Some Exact Solutions
- Plane Couette Flow
- Fully-developed Plane Poiseuille's Flow
- Plane Couette-Poiseuille's Flow

Topic No. 143

What is a Boundary Layer?

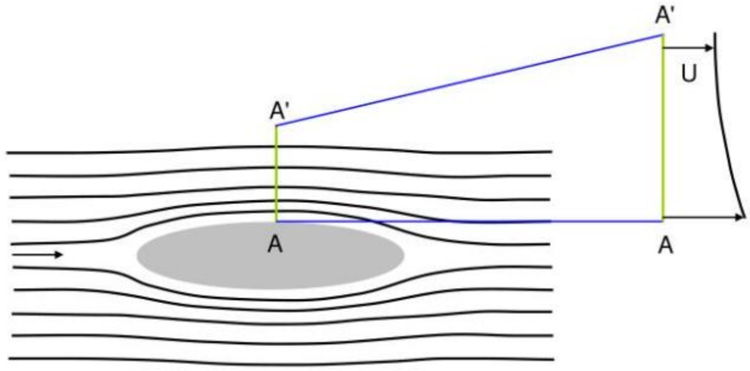
- A **boundary layer** is a layer of flowing fluid near a boundary where viscosity keeps the flow velocity close to that of the boundary.

- If the boundary is not moving, then the normal and tangential flow velocities at the boundary vanish, and the normal and tangential flow velocities near the boundary are small compared to those of the inviscid (potential) flow far from the boundary.
- Consider the **potential flow** around the body below. Constant rectilinear potential flow prevails far upstream and downstream of the body. The potential flow is forced to accelerate around the body. The potential flow field satisfies the condition of zero normal velocity at the boundary, but not zero tangential velocity at the boundary.



Potential Flow around the Body

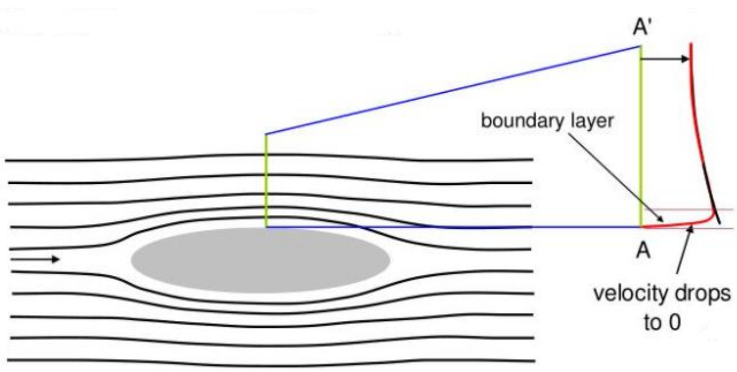
- The potential flow is forced to accelerate around the body. Thus $u(\partial u/\partial s) > 0$ along a streamline near the body from near the upstream stagnation point to section A−A'. The velocity profile along the line A−A' reflects this.



But Potential Flow is not the Whole Story

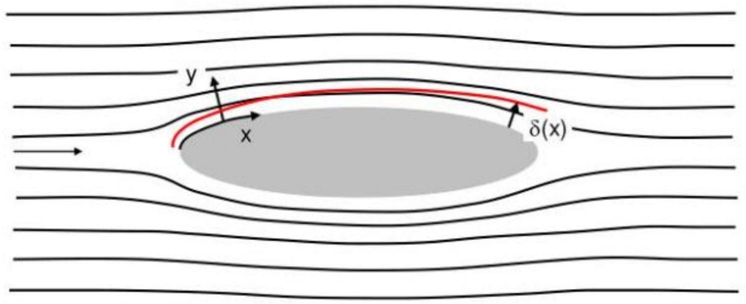
- If the body is not moving, there must be some region (perhaps very thin) where the flow velocity drops to zero. This region is called a **boundary layer**. We will find that boundary layer thickness depends on an appropriately defined **Reynolds number** of the flow, and as this number gets larger the boundary layer gets thinner.

- Here we are interested in the case of **large Reynolds number** (but not so large that the flow becomes turbulent).



Boundary-Attached Coordinates and Boundary Layer Thickness

- Let x denotes a boundary-attached streamwise coordinate, and y denote a boundary-attached normal coordinate, as noted below. Furthermore, let $\delta(x)$ be defined as some measure of **boundary layer thickness**, i.e. the thickness of the zone of retarded flow near the boundary (to be defined in more detail later), Then the boundary layer can be illustrated as follows (red line):



2D Navier-Stokes and Continuity Equations

- Since the x-y coordinate system is boundary-attached, it defines a curvilinear rather than Cartesian system. If the curvature is not too great, however, the governing equations for steady 2D flow around the body can be approximated as if in a Cartesian coordinate system:

$$u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}=-\frac{1}{\rho}\frac{\partial p_d}{\partial x}+\nu\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}\right)$$
$$u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}=-\frac{1}{\rho}\frac{\partial p_d}{\partial y}+\nu\left(\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}\right)$$
$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}=0$$

- Note that only the dynamic pressure p_d is used here. The total pressure p has been decomposed as $p = p_h + p_d$, and the static pressure p_h has been cancelled out against the gravitational terms.

The End

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